



Terrestrial Biodiversity, Plant and Animal Species Specialist Assessment

Proposed construction of Walmer Housing Development
on Erf 11305, Port Elizabeth, in the Eastern Cape Province

Prepared for:

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DECLARATION OF INDEPENDENCE

I, Merika Louw, as duly authorised representative of BlueLeaf Environmental (Pty) Ltd, hereby confirm my independence (as well as that of BlueLeaf) as a specialist and declare that neither I nor BlueLeaf have any interest, be it business, financial, personal or other, in any proposed activity, application or appeal in respect of which BlueLeaf Environmental (Pty) Ltd was appointed as environmental assessment practitioner in terms of the National Environmental Management Act, 1998 (Act No. 107 of 1998), other than fair remuneration for work performed, specifically in connection with the Specialist Assessments for the proposed construction of the Walmer Housing Development on Erf 11305, Port Elizabeth, in the Nelson Mandela Bay Metropolitan Municipality in the Eastern Cape Province. I further declare that I am confident in the results of the studies undertaken and conclusions drawn because of it – as is described in this report.

AUTHOR

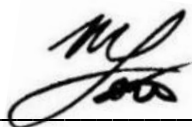
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Merika has over 15 years' experience in environmental consulting and botanical specialist services in South Africa. She holds a BSc Honours in Botany and an MSc in Restoration Ecology (Botany) from the Nelson Mandela University in Port Elizabeth. She has experience in environmental consulting – including Basic Assessments, Environmental Impact Assessments, Rehabilitation and Environmental Management Plans, and Environmental Auditing – as well as botanical specialist work – including Botanical and Vegetation Specialist Assessments and Ecological Sensitivity Assessments – for applications in different economic sectors. For the past six years, her focus has been on mining applications in the Eastern Cape. Her other experience includes residential, agricultural, road and service infrastructure developments, as well as environmental audits on residential, renewable energy and mining projects.

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Roy has over 19 years' experience in environmental consulting and specialist services in South Africa. Various projects throughout Africa have also been undertaken. Projects include baseline studies, impact assessments and compliance auditing for various large-scale projects including numerous wind farms, roads (National and Provincial), and infrastructure development projects. Blue Leaf also offers a wide range of in-house specialities including but not limited to Ecological and Botanical assessments, Biodiversity studies, Plant and Animal Search and Rescue, Fauna and Flora permits, Aquatic Assessments, Agricultural and Soil Assessments and Environmental and Venomous animals training workshops.

Roy holds a BSc Honours in Geology and an MSc in Botany from the Nelson Mandela University in Port Elizabeth. He is currently busy with his PhD (Doctorate degree) in Botany and Soil Science. He has over 19 years' experience in the environmental consulting focussing on Ecological and Agricultural Assessments, Geological and Geotechnical analysis, Environmental Management Plans, mining applications and various environmental impact studies.

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1. Introduction

BlueLeaf Environmental (Pty) Ltd (BlueLeaf) has been appointed by Abantu Environmental Services (Pty) Ltd to undertake a Terrestrial Biodiversity, Plant and Animal Species Specialist Assessment as part of the Scoping and Environmental Impact Assessment (EIA) process being conducted by Abantu Environmental Services on behalf of the Nelson Mandela Metropolitan Municipality (NMMM). The project involves the establishment of the Walmer Housing Development – to be located west of Walmer Township and Victoria Drive in Gqeberha.

This investigation is guided by the National Environmental Management Act (NEMA) (Act No. 107 of 1998) as well as the Department of Forestry, Fisheries and the Environment's (DFFE) online screening tool protocols for the following specialist themes:

1. Terrestrial Biodiversity
2. Plant Species
3. Animal Species.

1.1. Project locality

The proposed housing development will be located on Erf 11305, Port Elizabeth, in the Nelson Mandela Bay Metropolitan Municipality (Port Elizabeth/Gqeberha) in the Eastern Cape Province (Figure 1.1).

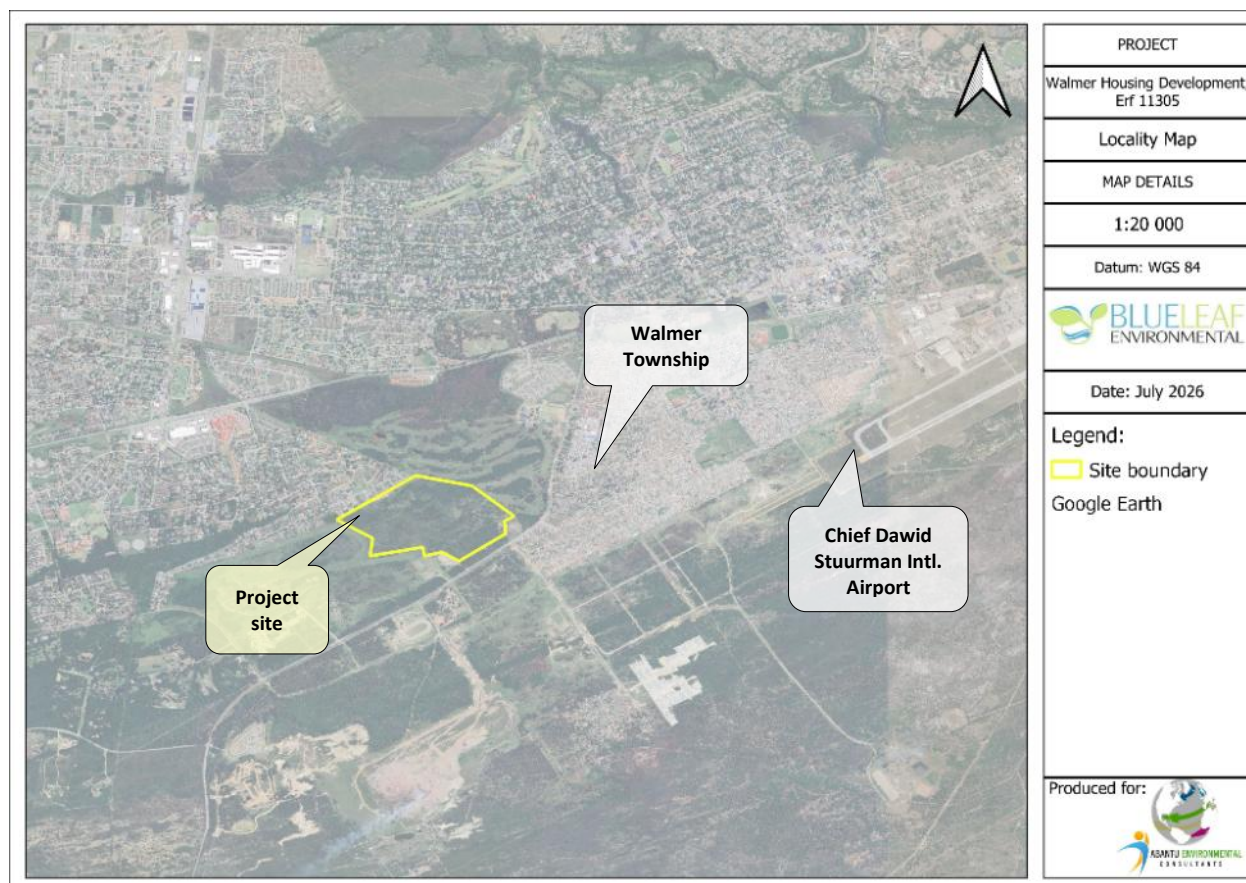


Figure 1.1: Location of the proposed Walmer housing development in Port Elizabeth.

The development site is located between the suburbs of Walmer Heights and Beaumont Estate, to the west and northwest; the old Arlington Racecourse, to the west; SmartStone Port Elizabeth, to the south; the Walmer Country Club, to the north and northeast; Walmer Township, to the east, and Victoria Drive (M18) and open space, to the southeast.

1.2. Project description

The project will involve construction of the Walmer Housing Development, including associated facilities and infrastructure. The project site is 43.73 hectares in total extent – of which 20.48 ha will remain as Open Space Zone 1 and 3 (open/grassed areas and forest/thicket patches, respectively); 13.53 ha will be developed into residential areas; 3.1 ha will be developed into business, community and authority zones, and 6.61 ha will be developed as roads/transport zones. See site layout plan shown in Figure 1.2.

The Gross Residential Area (single dwellings) is 11.11 hectares – with a Gross Residential Density of 14.9 units per hectare (single dwellings) and a Nett Residential Density of 58.69 units per hectare (single dwellings). The average erf size for single dwellings is 170.41 m². No internal service infrastructure is currently in place on site – however, electrical, sewage and water infrastructure will be connected to the existing bulk infrastructure currently servicing the surrounding residential areas.

The housing development will provide a mix of housing typologies and associated facilities which include the following:

Housing typologies and associated facilities:

- Free Basic House/RDP
- GAP/FLISP
- Social Housing
- Open/GAP market housing

Other land uses:

- Business Zone I: to provide for businesses such as a supermarket, bottle store, service trades etc. with parking provision at 3.33 bays per 100 m²
- Institutional Zone I: school site
- Authority Zone: authority uses such as police station, fire services, telecommunications facilities etc.
- Open Space Zone I: public open space.
- Open Space Zone 3: conservation usage.
- Special Use: conservation usage (forest patches)
- Transportation Zone II: public zone.

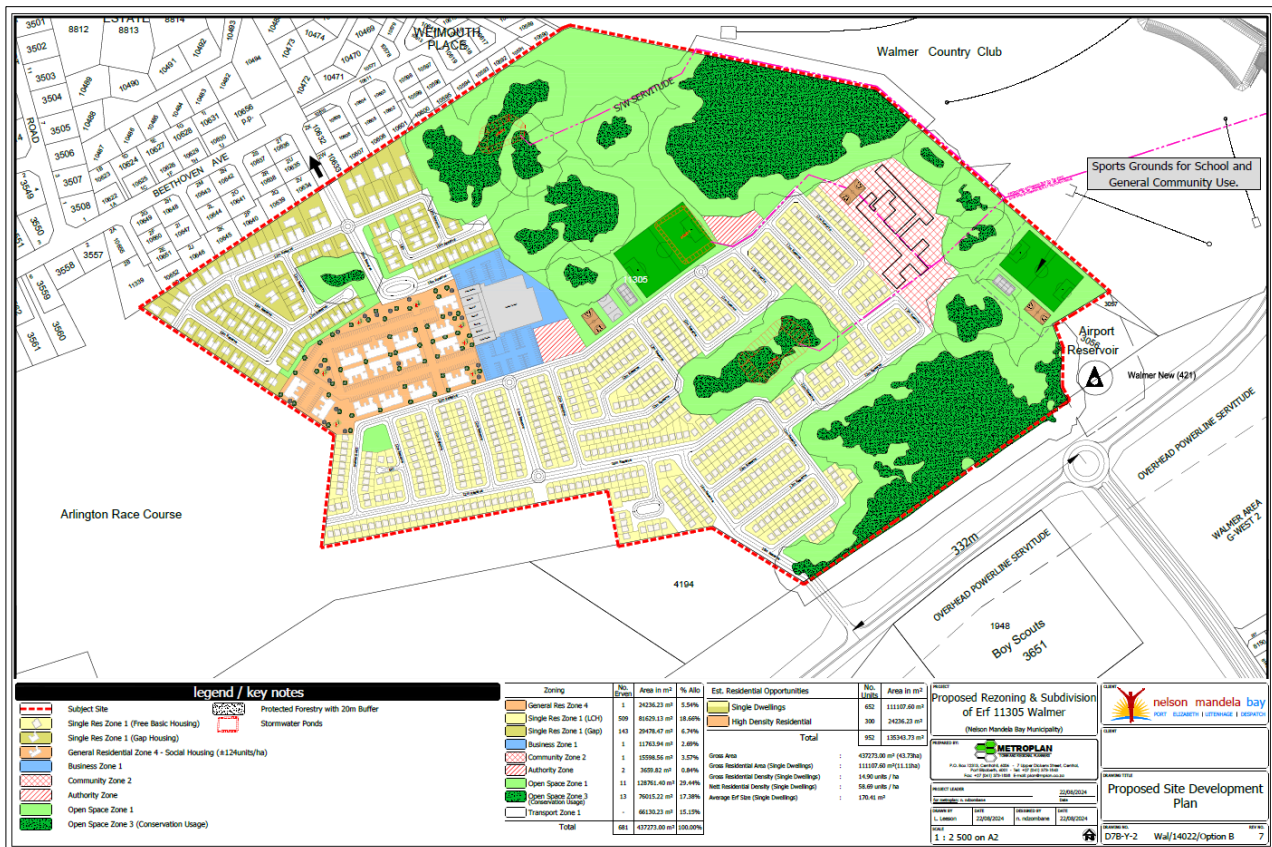


Figure 1.2: Site layout plan for the proposed housing development (Metroplan Town and Regional Planners, 2024).

1.3. Legislative context

The following legislation is directly relevant when assessing the ecological environment relating to the proposed Terrestrial Biodiversity, Plant and Animal Species Specialist Assessment.

National Environmental Management Act (NEMA) (Act No. 107 of 1998, as amended) and the Specialist Assessment Protocols (GNR 320 of 2020)

The National Web based Environmental Screening Tool (<https://screening.environment.gov.za/screeningtool/>) is a geographically based web-enabled application which allows a proponent intending to apply for environmental authorization in terms of the Environmental Impact Assessment (EIA) Regulations, 2014 (as amended), to screen their proposed site for any environmental sensitivity.

The Screening Tool also provides site specific EIA process and review information, for example, the Screening Tool may identify if an industrial development zone, minimum information requirement, Environmental Management Framework or bioregional plan applies to a specific area.

Further to this, the Screening Tool identifies related exclusions and/or specific requirements including specialist studies applicable to the proposed site and/or development, based on the national sector classification and the environmental sensitivity of the site.

Finally, the Screening Tool allows for the generating of a Screening Report referred to in Regulation 16(1)(v) of the Environmental Impact Assessment Regulations, 2014 (as amended), whereby a Screening Report is required to accompany any application for Environmental Authorization and as such the tool has been developed in a manner that is user friendly and no specific software or specialized GIS skills are required to operate this system.

Procedures for the assessment and minimum criteria for reporting on identified environmental themes in terms of Sections 24(5)(a), (h) and 44 of the National Environmental Management Act, 1998, when applying for Environmental Authorization have been Gazetted (GNR 320 of 20 March 2020). In terms of Sections 24(5)(a), (h) and 44 of the National Environmental Management Act, 1998. These procedures prescribe general requirements for undertaking site sensitivity verification and for protocols for the assessment and minimum report content requirements of environmental impacts for environmental themes for activities requiring environmental authorization, as contained in the Schedule therein. When the requirements of a protocol apply, the requirements of Appendix 6 of the EIA Regulations, 2014 (as amended), promulgated under Sections 24(5) and 44 of the National Environmental Management Act, 1998, are replaced by these requirements.

The contents of this specialist report comply with the legislated requirements as described in the following environmental theme and associated specialist assessment protocols as listed in the projects' Screening Tool Report:

- Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Terrestrial Biodiversity.
- Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Terrestrial Plant Species.
- Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Terrestrial Animal Species.

The Species Environmental Assessment Guidelines published in 2022 were used for the implementation of the Terrestrial Plant and Animal Species Protocols in this report.

National Environmental Management Act (NEMA) (Act No. 107 of 1998, as amended) and the EIA Regulations, 2014 (as amended)

Although the Specialist Assessment Protocols (as listed above) supersede this legislative requirement, the contents of this specialist report still complies with the legislated requirements as described in Appendix 6 of the National Environmental Management Act (Act. No 107 of 1998; NEMA): EIA Regulations (2014) and updated in 2017 (GNR 326 of 2017).

Other national legislation

Other national legislation relevant to this project includes:

Title of legislation or guideline	Administering authority	Applicability to the project
National Environmental Management Act (NEMA) (Act No. 107 of 1998):	DEDEAT	The activity triggers activities listed in NEMA EIA Regulations GN R. 320.

Title of legislation or guideline	Administering authority	Applicability to the project
Environmental Impact Assessment (EIA) Regulations, 2014, as amended		
National Water Act, 1998 (Act No. 36 of 1998)	Department of Water and Sanitation (DWS)	Infrastructure may impact on existing surface water systems. This impact is only mentioned in this report (if relevant) and NOT discussed in detail.
National Environment Management: Biodiversity Act (NEMBA), 2004 (Act No. 10 of 2004)	DEDEAT	The proposed development must: – Conserve endangered ecosystems and protect and promote biodiversity. – Assess the impacts of the proposed development on endangered ecosystems. – No protected species may be removed or damaged without a permit. – The proposed site must be cleared of alien vegetation using appropriate means.
National Forests Act, 1998 (Act No. 84 of 1998)	Provincial Department of Forestry	Requires that a permit be obtained should any forest vegetation or protected trees be removed during the construction phase of the project.

Relevant Provincial and Municipal legislation includes:

Title of legislation or guideline	Administering authority	Applicability to the project
Nature and Environmental Conservation Ordinance (No.19 of 1974)	DEDEAT	Listing of protected animal species. Permits are required for removal and relocating any of these listed species in the EC.
Eastern Cape Biodiversity Conservation Plan (2007, revised 2018)	DEDEAT	Listing all Freshwater Aquatic CBAs and ESAs within the study site.
Nelson Mandela Bay Bioregional Plan, 2015	DEDEAT, NMBM	The most fine-scale bioregional plan available for the NMBM, listing CBAs, ESAs, Critical Ecological Process Areas and Vegetation Types within the study site.
Nelson Mandela Bay Municipal Spatial Development Framework, 2025	DEDEAT, NMBM	Provides a spatial expression of the vision of the NMBM Integrated Development Plan (IDP): 2022/23 to 2026/27. Provides a guide to promote the wise demarcation and responsible allocation of land uses during land use planning.

1.4. Alternatives

No site alternatives or alternative layouts are proposed.

1.5. Public consultation

No consultation requirements were identified during the drafting of this specialist report. The findings of this report can be presented to stakeholders and I&APs as part of the EIA Public Participation Process (PPP). No comments were received to date on this report.

1.6. Objectives

The objectives of the project are listed below. These objectives are based on the requirements of each specialist protocol as listed in the DFFE Screening Report:

- Describe both the existing area as well as the area prior to construction in terms of its current ecological characteristics and the general sensitivity of these components to change.
- Confirm if there are any outright fatal flaws to the establishment of the proposal at its current location from a biodiversity perspective.
- Map all existing areas to be directly affected by the proposals in terms of its current and previous biodiversity sensitivity (constraints).
- Map all 'No-Go' areas.
- Describe the likely scope, scale, and significance of impacts (positive and negative) on biodiversity components of the area associated with the construction of the proposals.
- Make recommendations on the scope of any mitigation measures that may be applied during construction to avoid/reduce the significance of the identified construction-related impacts.
- Describe the likely scope, scale, and significance of impacts (positive or negative) on the biodiversity components associated with the operation or use of the proposals.
- Make recommendations on the scope of any mitigation measures that may be applied to avoid/reduce the significance of the operational phase-related impacts. These mitigation measures could also be design recommendations as well as operational controls, monitoring programmes, management procedures and the like.
- It will be particularly important to identify any rehabilitation measures that can be reasonably applied on the completion of the construction works.
- Broadly comment on the cumulative ecological impacts (positive or negative) associated with the construction and/or operation of the proposals.

It should be noted that only datasets and base data relevant to the study area and affected environmental features are discussed below.

1.7. Assumptions and limitations

- The report is based on currently available information and, as a result, limited by the information provided by the Client.
- The report is limited by seasonality as the presented data will be based on one site survey of plant and animal species conducted within a single season (winter) of a single year (2026).

1.8. Project Area of Influence

The Project Area of Influence (PAOI) is defined according to important ecosystem processes and functions that may be plausibly affected by the proposed development and its associated activities. The PAOI sets the minimum spatial extent of the study area, and the assessment will be focused within this area. The following site descriptors were used to delineate each PAOI (Figure 1.3).

PAOI	Area (ha)	Description	Probability of impact occurring
Primary PAOI	43.73	The Primary PAOI includes all proposed infrastructure within the boundary of the development site (houses, roads, service infrastructure, etc.). This is the area directly impacted by the proposed development and includes all permanent footprints.	Definite
Secondary PAOI	33.25	The secondary PAOI includes all areas within a 100 m buffer of the proposed development boundary. These areas are not directly impacted by the development unless temporary footprints like site camps, laydown areas and stockpiles are placed in them. Assessing this PAOI will not only result in identifying potential indirect and cumulative impacts but will also allow for micro-movement of infrastructure.	Likely
Total PAOI	76.98	The Primary and Secondary PAOI's are collectively referred to as the Total PAOI (or just the PAOI) or study site in this report and demarcates the extent of the site that will be assessed.	Likely

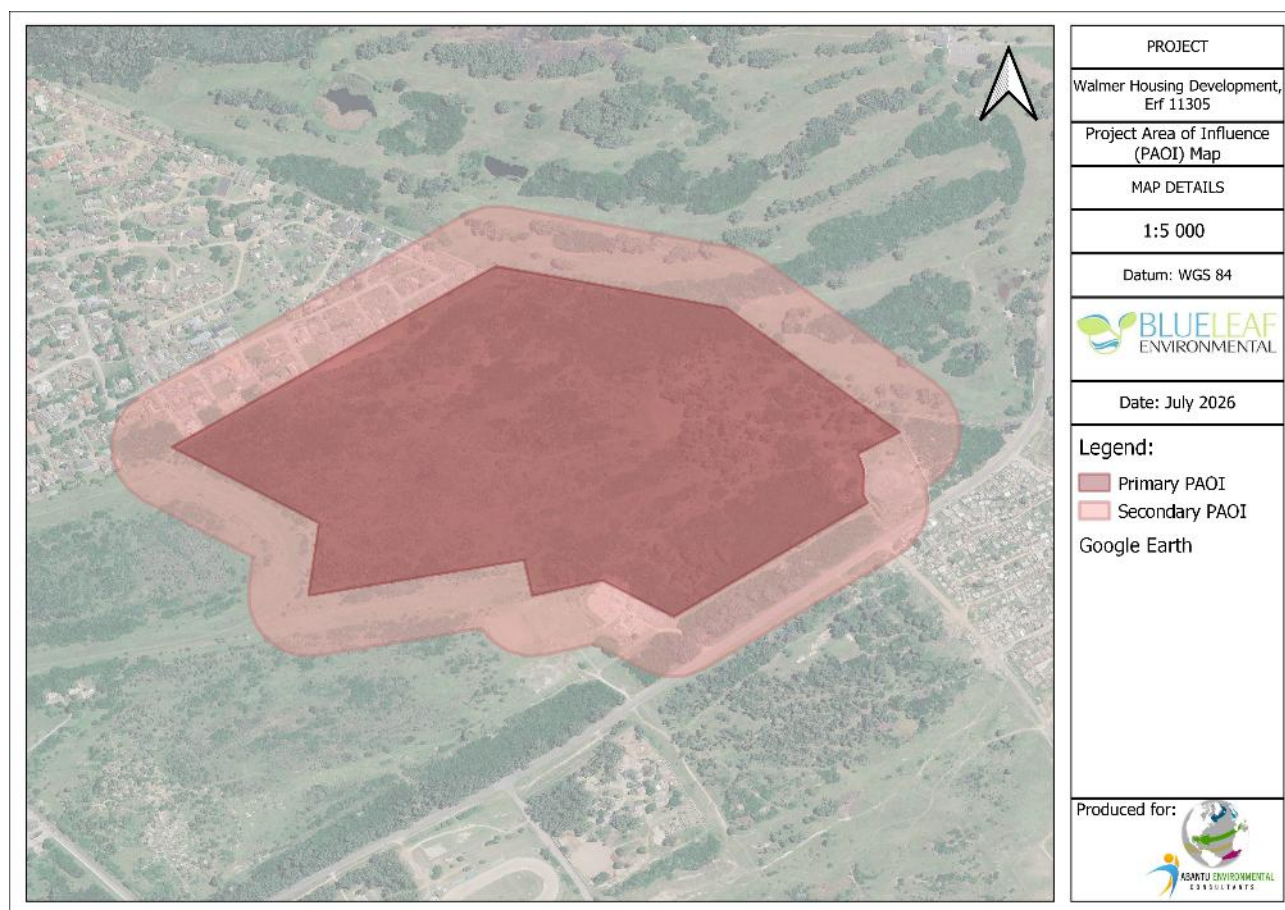


Figure 1.3: Site map showing the Primary and Secondary PAOI's for the proposed housing development.

2. Approach and Methodology

The aim of this assessment is to identify areas of ecological importance and to evaluate these in terms of their conservation importance. To do so, the ecological sensitivity of the area is assessed, as well as an identification of potential plant and animal Species of Conservation Concern (SCC) that may occur in habitats present in the area. The condition and sensitivity of the vegetation, habitat and connectivity, as well as the presence of plant and animal SCC, will determine biodiversity sensitivities for the study site.

The study site was assessed using a two-phased approach. Firstly, a desktop assessment of the site was conducted in terms of current biodiversity programmes and plans.

Further to the above, a site visit was conducted on 8 July 2026, in winter. Plant species that are not visible (not flowering or latent/dormant below ground during this time of year) or animal species that were not present on site, at this specific time of year, may have been overlooked. The site visit served to inform the type of impacts of the proposed development and how significantly the development would impact on the surrounding terrestrial environment.

2.1. Plant species classification

To identify plant species that potentially occur naturally in the project area, firstly required an understanding of the broad botanical habitats in the area. These habitats were identified according to various ecological characteristics, including vegetation type (NMBM Bioregional Plan, 2015; SANBI VegMap, 2024); the degree of transformation of the vegetation; geology and soil type, and topography.

The potential occurrence of plants that are considered Species of Conservation Concern (SCC) within the project area was identified through a detailed assessment of the site. Published literature and online resources that are continuously updated with new species observations were consulted to compile lists of plant, including:

- The Vegetation of South Africa, Lesotho and Swaziland (Mucina and Rutherford, 2006).
- National VegMap, 2024.
- NMBM Bioregional Plan, 2015.
- iNaturalist online database ([inaturalist.org](https://www.inaturalist.org)).
- Plants of South Africa/New POSA online database (posa.sanbi.org).
- SANBI Red List of South African Plants online database.
- National Web-based Screening Tool (screening.environment.gov.za), and
- Global Biodiversity Information Facility online database (gbif.org).

Plant Species of Conservation Concern (SCC) were limited to endemic flora, and were defined to include:

- Red Data species identified using the IUCN Red List of Threatened Species.

- Red Data species identified using the Red List of South African Species. This includes all threatened species that are assessed according to the IUCN Red List Criteria as Critically Endangered, Endangered, Vulnerable, Near Threatened, Rare, Extremely Rare, or Data Deficient.
- Plants listed in terms of Section 56 of the National Environmental Management: Biodiversity Act, 2004 (NEMBA) (Act No. 10 of 2004, as amended), and regulated by the Threatened or Protected Species (TOPS) Regulations, 2007. This includes threatened species that are Critically Endangered, Endangered, Vulnerable and Protected.
- All tree species listed by the National Forests Act (Act No. 84 of 1998; NFA) – List of all tree species which are protected under Section 12 (published January 2023) – as protected by law.

In addition to plant SCC, the following plants were also identified:

- Plants protected by the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).
- Plants listed in terms of the Nature and Environmental Conservation Ordinance (NECO) (No. 19 of 1974).

The inclusion of plant species on CITES Appendices (I–III) and NECO Schedules (1–2) are not necessarily equivalent to the species' conservation status. Many common species are pragmatically included on these lists even though their conservation status may not be of demonstrated concern. A permit is however required for the removal of species that occur on CITES and NECO lists.

Several sensitive plant species, identified by the Department of Forestry, Fisheries and the Environment's (DFFE) National Web-based Environmental Screening Tool as important, required specific consideration. These species include:

Sensitivity	Feature(s)
High	Agathosma stenopetala
Low	Low Sensitivity
Medium	Sensitive species 1252
Medium	Argyrobium crassifolium
Medium	Aspalathus recurvospina
Medium	Sensitive species 991
Medium	Lotononis acuminata
Medium	Selago rotundifolia
Medium	Erica chloroloma
Medium	Erica zeyheriana
Medium	Gymnosporia elliptica
Medium	Sensitive species 588
Medium	Sensitive species 657

Medium	Sensitive species 670
Medium	<i>Centella tridentata</i> var. <i>hermanniifolia</i>
Medium	<i>Rapanea gilliana</i>
Medium	<i>Holothrix longicornu</i>
Medium	<i>Agathosma gonaquensis</i>
Medium	<i>Agathosma stenopetala</i>
Medium	<i>Corpuscularia lehmannii</i>
Medium	<i>Caputia scaposa</i> var. <i>addoensis</i>
Medium	Sensitive species 448
Medium	<i>Erica glumiflora</i>
Medium	Sensitive species 654

Where only a sensitive plant unique number is provided instead of a name the species may be prone to illegal harvesting and must be protected. The name will therefore not be made public.

The following criteria were evaluated during the site visit to assess habitat suitability for these species:

1. Vegetation type and cover.
2. Geology and soil type.
3. Rock cover, and
4. Topography.

Sensitive habitats were identified as those habitats that are vulnerable to disturbances and supporting SCC in the project area.

A visit to the project area was conducted on 27 July 2026 in order to:

1. Assess the micro-positioning for infrastructure.
2. Confirm the occurrence of sensitive vegetation habitats.
3. Note any occurrence of plant SCC and other indicator species occurrences, and
4. Assess the extent of current threats (not project related) on vegetation and plant SCC (e.g. evidence for direct exploitation, habitat transformation, etc.).

2.2. Animal species classification

The potential occurrence of fauna (amphibians, reptiles, mammals, birds, and butterflies) in the vicinity of the project area was determined according to the habitat characteristics of the area and the species' habitat requirements. Published literature and online resources that are continuously updated with new species observations were consulted to compile lists of fauna, including:

- Du Preez & Carruthers (2017), Frog Atlas of Southern Africa (FrogMap) for amphibians.
- Alexander & Marais (2007), Reptile Atlas of Southern Africa (ReptileMap1) for reptiles.
- Skinner & Chimimba (2005), Mammal Atlas of Southern Africa (MammalMap1) for mammals.
- Southern African Bird Atlas Project 2 (SABAP2) for birds (Harrison et al. 1997).
- Woodhall (2020), Atlas of African Lepidoptera (LepiMap1) for butterflies.
- iNaturalist.

- Global Biodiversity Information Facility (GBIF).

Faunal Species of Conservation Concern (SCC) were defined to include:

- Fauna with their distribution ranges limited to the Eastern Cape Province.
- Red Data species identified using the IUCN Red List of Threatened Species.
- Red Data species identified using the Red List of South African Species. Includes all species that are assessed according to the IUCN Red List Criteria as Critically Endangered, Endangered, Vulnerable, Near Threatened, Rare, Extremely Rare or Data Deficient. Listings were corroborated with data from the South African amphibian (Measey, 2011), reptile (Bates *et al.*, 2013), mammal (Child *et al.*, 2016), bird (Taylor *et al.*, 2015), and butterfly (Mecenero *et al.*, 2020) conservation assessments.
- Species listed in terms of Section 56 of the National Environmental Management: Biodiversity Act, 2004 (NEMBA) (Act No. 10 of 2004, as amended), and regulated by the Threatened or Protected Species (TOPS) Regulations, 2007. Includes species that are Critically Endangered, Endangered, Vulnerable and Protected.

In addition to Faunal SCC, the assessment also identified:

- Species protected by the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).
- Species listed in terms of the Nature and Environmental Conservation Ordinance (NECO) (No. 19 of 1974).

The inclusion of faunal species on CITES Appendices (I–III) and NECO Schedules (1–2) are not necessarily equivalent to the species' conservation status. Many common species are pragmatically included on these lists even though their conservation status may not be of demonstrated concern. A permit is required for the removal of species that occur on CITES and NECO lists.

Expected Faunal SCC with very low detection probabilities were assessed according to the following criteria:

1. Probability of occurrence in the project area (4 categories: confirmed [observed during the site visit], high, medium, and low),
2. Potential distribution within the project area, and
3. Current threats (not project related).

Several sensitive animal species, identified by the Department of Forestry, Fisheries and the Environment's (DFFE) National Web-based Environmental Screening Tool as important, required specific consideration. These species include:

Sensitivity	Feature(s)
High	Aves-Circus ranivorus
High	Aves-Stephanoaetus coronatus
High	Aves-Neotis denhami
High	Aves-Bradypterus sylvaticus
Medium	Aves-Stephanoaetus coronatus
Medium	Aves-Neotis denhami
Medium	Aves-Eupodotis senegalensis
Medium	Insecta-Chrysoritis thysbe whitei
Medium	Mammalia-Chlorotalpa duthieae
Medium	Sensitive species 8
Medium	Invertebrate-Aneuryphymus montanus

Where only a sensitive animal unique number is provided instead of a name the species may be prone to illegal harvesting and must be protected. The name will therefore not be made public.

Because the likelihood of detecting animal SCC during environmental authorisation assessments is extremely low (even with optimal search methods and during optimal seasonal sampling) a precautionary principle was applied in the following way when assessing habitat suitability for species in the project area:

- Potentially suitable habitat exists in proximity of known locations for the species, then the species is assigned a high probability of occurrence and assumed to be present.

The following criteria were evaluated during the site visit to assess habitat suitability for these species:

1. Vegetation type and cover,
2. Presence of host plants (in the case of butterflies),
3. Geology and soil type,
4. Rock cover,
5. Topography,
6. Habitat disturbance, and
7. Habitat connectivity.

2.3. Biodiversity classification

The Nelson Mandela Bay Municipality Bioregional Plan (2015) is the most fine-scale and detailed bioregional or conservation plan available for the Nelson Mandela Bay Municipality, detailing the presence and layout of Critical Biodiversity Areas, Ecological Support Areas, Ecological Process Areas, Land Cover and Vegetation Types.

Critical Biodiversity Areas (CBAs) are features critical for the conservation of biodiversity and maintenance of ecosystem functioning and should remain in a natural state as far as possible. CBAs also include freshwater components. To assist in the development of these CBAs, each planning unit was classified by C-Plan and Marxan based on a combination of factors including vegetation type, connectivity, habitat condition and presence of Red List Threatened species.

All features were grouped into the following CBA categories as listed in the Nelson Mandela Bay Bioregional Plan:

Category on the CBA Map	Description	Land Management Objective	Land Management Recommendations	General Land-Use
Protected Areas (PA 1 & 2)	Formal Protected Areas (PA1) and Protected Areas pending declaration (PA2), in terms of the NEM: Protected Areas Act and ECA.	Maintain or restore NATURAL structure and ecosystem functioning.	Maintain formal conservation protection.	Conservation (non-consumptive).
Critical Biodiversity Areas (CBAs)	ANY land or aquatic site required to meet biodiversity pattern (features), process (ecological), or species targets.	Maintain or restore NATURAL structure and ecosystem functioning.	Obtain formal conservation protection for highest priority sites and institute conservation management for all sites.	Conservation (non-consumptive).
Ecological Support Area 1 (ESA1)	Agricultural or partly degraded land is essential for connectivity in the landscape, particularly between CBAs.	Maintain existing extensive land-uses (e.g. sustainable livestock grazing) where these facilitate ecological functioning and connectivity between adjacent CBAs; or restore NATURAL structure and ecosystem functioning; or rehabilitate to near natural state to facilitate connectivity between adjacent CBAs.	Maintain existing extensive land-uses to facilitate ecological connectivity or manage for conservation.	Extensive agriculture or similar low intensity purposes that promote conservation and ecological connectivity (sustainable consumptive or non-consumptive).

2.4. Vegetation mapping

Mucina and Rutherford (2006) developed the National Vegetation Map (VegMap) – first published as a print book in 2006 and updated and published as a CD-ROM set in 2010. The further updated VegMap (2018) was published under Dayaram *et al.* (2019). The ecosystem threat status of these vegetation types, based on the Mucina and Rutherford (2006) and Dayaram *et al.* (2019) vegetation types, were gazetted in 2022 (NEMBA Revised National List of Ecosystems that are Threatened and in need of Protection, G. NR. 2747, 18 November 2022).

The NEMBA Revised National of Ecosystems that are Threatened and in need of protection (G. NR. 2747 of 2022) contains a national list of threatened terrestrial ecosystems. This is the latest gazetted list of threatened ecosystems available for the country.

The latest updated vegetation classification and map is the VegMap (2024), available as a dataset on the SANBI website (sanbi.org.). This update describes 465 national vegetation types in detail,

along with the most important species including endemic species and those that are biogeographically important, that provided the main basis for delineating terrestrial ecosystem types in the National Biodiversity Assessment (2025; <https://nba.sanbi.org.za>). However, the latest The VegMap (2024) classification of vegetation types has not yet been published or gazetted.

The NMBM Final Conservation Assessment and Plan (2010), which was incorporated into the NMBM Final Bioregional Plan (2015), provides a municipal-level vegetation classification and map for the Nelson Mandela Bay Municipality – which includes the proposed development area.

2.5. Protected areas

The National Environmental Management Protected Areas Act (NEMPAA) (Act No. 57 of 2003) was developed to provide for the protection and conservation of ecologically viable areas representative of South Africa's biological diversity and its natural landscapes and seascapes. Refer to Section 3.9.5 for more detail on this matter.

2.6. Site Ecological Importance

The Screening Tool has identified species and ecosystem spatial triggers likely to indicate environmental sensitivity associated with a particular proposed development site, which in turn determined the necessity and requirements for conducting Specialist Assessments. The Species Environmental Assessment Guideline (SANBI, 2022) provided methodology for the identification of environmental sensitive areas within the PAOI.

Site Ecological Importance (SEI) is a function of the biodiversity importance (BI) of the identified receptor (e.g. species of conservation concern, the vegetation/fauna community or habitat type present on the site) and its resilience to impacts (receptor resilience [RR]):

$$\text{SEI} = \text{BI} + \text{RR}$$

BI in turn is a function of conservation importance (CI) and the functional integrity (FI) of the receptor:

$$\text{BI} = \text{CI} + \text{FI}$$

Conservation Importance

CI is defined as '*The importance of a site for supporting biodiversity features of conservation concern present, rare species, range-restricted species, globally significant populations of congregatory species, and areas of threatened ecosystem types, through predominantly natural processes.*' These criteria are defined as follow:

- **IUCN threatened and Near Threatened species** (CR, EN, VU and NT). Where the global (IUCN) and national assessments (SANBI Red List) differ for the same taxon, the national evaluation of status should be used in calculating SEI unless the global assessment is both more recent and of a more threatened category.

- **Rare species** are those included on South Africa's National Red List as Rare or Critically Rare or Extremely Rare.
- **Range-restricted species** – the presence of terrestrial flora, vertebrate and invertebrate fauna with a global population extent of occurrence (EOO) of 10 000 km² or less.
- **Globally significant populations of congregatory species** – a roughly estimated proportion (%) of the global population of a fauna species that congregate for breeding/feeding/hibernation/other reasons.
- **Significant areas of threatened vegetation types** – this is a function of both the area (size) being considered in relation to the total extent of that vegetation type (i.e. proportion) and how threatened (CR, EN, VU) the vegetation types are.
- **Natural processes** – natural unmanaged areas with low levels of ecological disturbance have largely intact natural processes such as pollination, seed dispersal and migration, and thus have greater intrinsic conservation importance than those that are modified through ecological disturbance.

While most of the features in the CI will be provided by the screening tool, it is important to note that CI is evaluated at a much finer spatial scale and based on fieldwork data collection and comprehensive desktop analyses performed during the site visit. CI is determined for each identified habitat/vegetation unit within the entire PAOI.

The following table was used to determine CI for each receptor:

Table 2.1: Conservation importance (CI) criteria.

CI	Criteria
Very high	➤ Confirmed or highly likely occurrence of CR, EN, VU or Extremely Rare or Critically Rare species that have a global EOO of < 10 km². ➤ Any area of natural habitat of a CR ecosystem type or large area (> 0.1% of the total ecosystem type extent) of natural habitat of EN ecosystem type. ➤ Globally significant populations of congregatory species (> 10% of global population).
High	➤ Confirmed or highly likely occurrence of CR, EN, VU species that have a global EOO of > 10 km². Include if there are less than 10 locations or < 10 000 mature individuals remaining. ➤ Small area (> 0.01% but < 0.1% of the total ecosystem type extent) of natural habitat of EN ecosystem type or large area (> 0.1%) of natural habitat of VU ecosystem type. ➤ Presence of Rare species. ➤ Globally significant populations of congregatory species (> 1% but < 10% of global population).
Medium	➤ Confirmed or highly likely occurrence of populations of NT species, threatened species (CR, EN, VU) and which have more than 10 locations or more than 10 000 mature individuals. ➤ Any area of natural habitat of threatened ecosystem type with status of VU. ➤ Presence of range-restricted species. ➤ > 50% of receptor contains natural habitat with potential to support SCC.
Low	➤ No confirmed or highly likely populations of SCC. ➤ No confirmed or highly likely populations of range-restricted species. ➤ < 50% of receptor contains natural habitat with limited potential to support SCC.
Very low	➤ No confirmed and highly unlikely populations of SCC. ➤ No confirmed and highly unlikely populations of range-restricted species. ➤ No natural habitat remaining.

Functional Integrity

FI is defined as “A measure of the ecological condition of the impact receptor as determined by its remaining intact and functional area, its connectivity to other natural areas and the degree of current persistent ecological impacts.” These criteria are defined as follow:

Connectivity to other natural areas – connectivity, which can also be measured conversely as the degree of habitat fragmentation, refers to how connected habitat patches are to each other, which has a significant influence on numerous ecological processes, such as migration and dispersal opportunities of biota and therefore genetic exchange between populations. Connectivity to other similar habitats becomes more important as the remaining intact and functional area of a habitat decreases, mainly because population sizes decrease and are therefore at greater risk from ecological perturbations and inbreeding effects. The degree of connectivity between habitat patches varies greatly with the dispersal ability of the taxon or taxon group (e.g. fossorial reptiles) in question.

Degree of current persistent negative ecological impacts – persistent negative impacts such as uncontrolled spread of alien and invasive flora effectively decreases both the remaining intact area and ecosystem functioning of a particular habitat.

Remaining intact and functional area – the proportion of the receptor that supports natural habitat with intact ecological processes – small areas are less likely to withstand ecological degradation compared to large areas, and the latter are therefore better able to maintain structure and function allowing for intact ecological processes. Ecological processes can be mostly intact and functional if the receptor area has low levels of current ecological disruptors, has good connectivity to other areas and is a relatively large area.

The following table was used to determine FI for each receptor:

Table 2.2: Functionality integrity (FI) criteria.

GI	Criteria
Very high	➤ Very large (> 100 ha) intact area for any conservation status of ecosystem type or > 5 ha for CR ecosystem types. ➤ High habitat connectivity serving as functional ecological corridors, limited road network between intact habitat patches. ➤ No or minimal current negative ecological impacts with no signs of major past disturbance (e.g. ploughing).
High	➤ Large (> 20 ha but < 100 ha) intact area for any conservation status of ecosystem type or > 10 ha for EN ecosystem types. ➤ Good habitat connectivity with potentially functional ecological corridors and a regularly used road network between intact habitat patches. ➤ Only minor current negative ecological impacts (e.g. few livestock utilising area) with no signs of major past disturbance (e.g. ploughing) and good rehabilitation potential.
Medium	➤ Medium (> 5 ha but < 20 ha) semi-intact area for any conservation status of ecosystem type or > 20 ha for VU ecosystem types. ➤ Only narrow corridors of good habitat connectivity or larger areas of poor habitat connectivity and a busy used road network between intact habitat patches. ➤ Mostly minor current negative ecological impacts with some major impacts (e.g. established population of alien and invasive flora) and a few signs of minor past disturbance. Moderate rehabilitation potential.
Low	➤ Small (> 1 ha but < 5 ha) area. ➤ Almost no habitat connectivity but migrations still possible across some modified or degraded natural habitat and a very busy used road network surrounds the area. Low rehabilitation potential.

GI	Criteria
	➤ Several minor and major current negative ecological impacts.
Very low	➤ Very small (< 1 ha) area. ➤ No habitat connectivity except for flying species or flora with wind-dispersed seeds. ➤ Several major current negative ecological impacts.

Biodiversity Importance

Recalling that BI is a function of CI and the FI of a receptor. BI can be derived from the following matrix of CI and FI:

Table 2.3: Biodiversity importance (BI) matrix.

Biodiversity Integrity (BI)		Conservation Importance (CI)				
Functional integrity (FI)	Very high	Very high	High	Medium	Low	Very low
	Very high	Very high	Very High	High	Medium	Low
	High	Very high	High	Medium	Medium	Low
	Medium	High	Medium	Medium	Low	Very low
	Low	Medium	Medium	Low	Low	Very low
	Very low	Medium	Low	Very low	Very low	Very low

Receptor Resilience

RR is defined as *“The intrinsic capacity of the receptor to resist major damage from disturbance and/or to recover to its original state with limited or no human intervention.”*

The fulfilling criteria to evaluate RR are based on the estimated recovery time required to restore an appreciable portion of functionality to the receptor and require justification. RR will be evaluated and justified for each evaluation based on the criteria below.

Table 2.4: Receptor resilience (RR) criteria.

RR	Criteria
Very high	➤ Habitat that can recover rapidly (~ less than 5 years) to restore > 75% of the original species composition and functionality of the receptor functionality, or species that have a very high likelihood of remaining at a site even when a disturbance or impact is occurring, or species that have a very high likelihood of returning to a site once the disturbance or impact has been removed.
High	➤ Habitat that can recover relatively quickly (~ 5–10 years) to restore > 75% of the original species composition and functionality of the receptor functionality, or species that have a high likelihood of remaining at a site even when a disturbance or impact is occurring, or species that have a high likelihood of returning to a site once the disturbance or impact has been removed.
Medium	➤ Will recover slowly (~ more than 10 years) to restore > 75% of the original species composition and functionality of the receptor functionality, or species that have a moderate likelihood of remaining at a site even when a disturbance or impact is occurring, or species that have a moderate likelihood of returning to a site once the disturbance or impact has been removed.
Low	➤ Habitat that is unlikely to be able to recover fully after a relatively long period: > 15 years required to restore ~ less than 50% of the original species composition and functionality of the receptor functionality, or species that have a low likelihood of remaining at a site even when a disturbance or impact is occurring, or species that have a low likelihood of returning to a site once the disturbance or impact has been removed.
Very low	➤ Habitat that is unable to recover from major impacts, or species that are unlikely to remain at a site even when a disturbance or impact is occurring, or species that are unlikely to return to a site once the disturbance or impact has been removed.

After evaluation of both BI and RR as described above, it is possible to evaluate SEI from the final matrix in Figure 2.3.

SEI description

The SEI was described in the above manner for each impact receptor identified within the PAOI and mapped in Section 5 of this report. The guidelines below (Figure 2.5) were used to interpret the SEI outcomes:

Table 2.5: Guidelines for interpreting SEI in the context of the proposed development activities

SEI	Interpretation in relation to proposed development activities
Very high	➤ Avoidance mitigation – no destructive development activities should be considered. Offset mitigation not acceptable/not possible (i.e. last remaining populations of species, last remaining good condition patches of ecosystems/unique species assemblages). Destructive impacts for species/ecosystems where persistence target remains.
High	➤ Avoidance mitigation wherever possible. Minimisation mitigation – changes to project infrastructure design to limit the amount of habitat impacted; limited development activities of low impact acceptable. Offset mitigation may be required for high impact activities.
Medium	➤ Minimisation and restoration mitigation – development activities of medium impact acceptable followed by appropriate restoration activities
Low	➤ Minimisation and restoration mitigation – development activities of medium to high impact acceptable followed by appropriate restoration activities
Very low	➤ Minimisation mitigation – development activities of medium to high impact acceptable and restoration activities may not be required

2.7. Impact assessment

The impacts that may result from the planning and design phase, construction phase and operational phase of the proposed development were assessed according to several criteria to arrive at an overall significance rating. The criteria used were as follows (based on DEAT 2002 - Impact Significance, IEM Information Series 5; and DEAT 2006 - Assessment of Alternatives and Impacts in support of the EIA Regulations, IEM Guideline Series 5):

Table 2.6: Criteria used in determining significance ratings to potential impacts.

ASPECT	IMPACT RATING
Status of the Impact	A statement of whether the impact is positive (a benefit), negative (a cost), or neutral
Direct impact	Impacts that are caused directly by the activity and generally occur at the same time and at the place of the activity. These impacts are usually associated with the construction, operation or maintenance of an activity and are generally obvious and quantifiable.
Indirect Impacts	Indirect impacts are not a direct result of the project but are often produced away from or because of a complex impact pathway related to the project.
Cumulative Impacts	Impacts that result from the incremental impact of the proposed activity on a common resource when added to the impacts of the past, present, or reasonably foreseeable future activities. Cumulative impacts can occur from the collective impacts of individual minor actions over a period of time and can include both direct and indirect impacts.
Nature of the Impact	The evaluation of the nature is impact specific. Most negative impacts will remain negative, however, after mitigation, significance should reduce to: ➤ Positive ➤ Negative

ASPECT		IMPACT RATING		
Extent	A description of whether the impact would occur on a scale limited to within the study area (local), limited to within 5 km of the study area (area) on a regional scale. i.e. the Kouga Municipality & Eastern Cape (Region); or would occur on a national or international scale.	Local	1	
		Area	2	
		Regional	3	
		National	4	
		International	5	
Duration	A prediction of whether the duration of the impact would be immediate and once-off (less than one month), more than once, but short term (less than one year), regular, medium term (1 to 5 years), long term (6 to 15 years), project life/permanent (> 15 years, with the impact ceasing after the operational life of the development or should be considered as permanent).	Immediate	1	
		Short term	2	
		Medium term	3	
		Long term	4	
		Project life/permanent	5	
Intensity	This provides an order of magnitude of whether or not the intensity (magnitude/size/frequency) of the impact would be negligible, low, medium, high or very high. This is based on the following aspects: ➤ An assessment of the reversibility of the impact (permanent loss of resources, or impact is reversible after project life; ➤ Whether or not the aspect is controversial; ➤ An assessment of the irreplaceability of the resource loss cause by the activity (whether the project will destroy the resources which are easily replaceable, or the project will destroy the resources which are irreplaceable and cannot be replaced; ➤ The level of alteration to the natural system, processes or systems.	Negligible	The impact does not affect physical, biophysical or socio-economic functions and processes.	1
		Low/potential harmful	The impact has limited impacts on physical, biophysical or socio-economic functions and processes.	2
		Medium/slightly harmful	The impact has an effect on physical, biophysical or socio-economic functions and processes, but in such a way that these processes can still continue to function albeit in a modified fashion.	3
		High/harmful	Where the physical, biophysical or socio-economic functions and processes are impacted on in such a way as to cause them to temporarily or permanently cease.	4
		Very high/disastrous	Where the physical, biophysical or socio-economic functions and processes are highly impacted on in such a way as to cause them to permanently cease.	5
Severity (extent + duration + intensity)				
Frequency	This provides a description of any repetitive, continuous or time-linked characteristics of the impact: Once-off (occurring any time during construction or operation); intermittent (occurring from time to time, without specific periodicity); periodic (occurring at more or less regular intervals); continuous (without interruption).	Once-off	Once	1
		Rare	1/5 to 1/10 years	2
		Frequent	Once a year	3
		Very frequent	Once a month	4
		Continuous	> once a day/per shift	5

ASPECT	IMPACT RATING		
Probability of occurrence	A description of the chance that consequences of that selected level of severity could occur during the exposure.		
	Highly unlikely	The probability of the impact occurring is highly unlikely due to its design or historic experience.	1
	Improbable	The probability of the impact occurring is low due to its design or historic experience.	2
	Probable	There is a distinct probability of the impact occurring.	3
	Almost certain	It is most likely that the impact will occur	4
	Definite	The impact will occur regardless of any prevention measures.	5
Incidence (frequency + probability)			
Risk rating	The risk rating is calculated based on input from the above assessments. The incidence of occurrence is calculated by adding the extent of the impact to the duration of the impact. The severity of the impact is calculated based on input from the extent of the impact, the duration and the intensity. Risk = Severity (extent + duration + intensity) x Incidence (frequency + probability) Significance: The significance of the risk based in the identified impacts has been expressed qualitatively as follows: ➤ Low – the impact is of little importance/insignificant but may/may not require minimal management. ➤ Medium – the impact is important; management is required to reduce negative impacts to acceptable levels. ➤ High – the impact is of great importance, negative impacts could render development options or the entire project unacceptable if they cannot reduce to acceptable levels and/or if they are not balanced by significant positive impact, management of impacts is essential.		

Based on a synthesis or combination of the information contained in the above-described criteria, and drawing on legal policies and guidelines as well as the status of the impacts and potential risks, the overall significance was determined as follows:

Table 2.7: Definition of significance ratings (positive and negative)

Significance	Description
Very high (VH) (150+)	An impact of very high significance will mean that the project cannot proceed, and that impacts are irreversible, regardless of available mitigation options.
High (H) (101-149)	An impact of high significance which could influence a decision about whether to proceed with the proposed project, regardless of available mitigation options.
Medium (M) (51-100)	If left unmanaged, an impact of medium-high significance could influence a decision about whether to proceed with a proposed project. An impact of Low-medium significance would have some effect during decision making about whether to proceed with a proposed project, however, mitigation for this type of impact is manageable.
Low (L) (25-50)	An impact of low significance would have little effect on decision making and only a small influence on project design or alternative motivation.
Very low (VL) (1-24)	An impact of very low significance is likely to contribute to positive decisions about whether to proceed with the project. It will have little effect and is unlikely to have an influence on project design or alternative motivation.
Negligible / zero impact	There will be no impact, or any impact identified can be viewed as negligible. This rating will be unlikely to have an influence on project design or alternative motivation.
Positive impact (+)	A positive impact is likely to result in a positive consequence/effect and is likely to contribute to positive decisions about whether to proceed with the project.

3. Site assessment

This chapter compares baseline information with field survey data collected. Data collected during the site visit was then compared to existing literature on the site which included vegetation classifications and biodiversity programmes and plans.

3.1. Site Sensitivity Verification

The DFFE Screening Tool Report indicated the following relevant sensitivities for the site:

Plant Species Theme:

The proposed development area environmental sensitivity for the plant species theme is **identified as high sensitivity** (Figure 3.1).

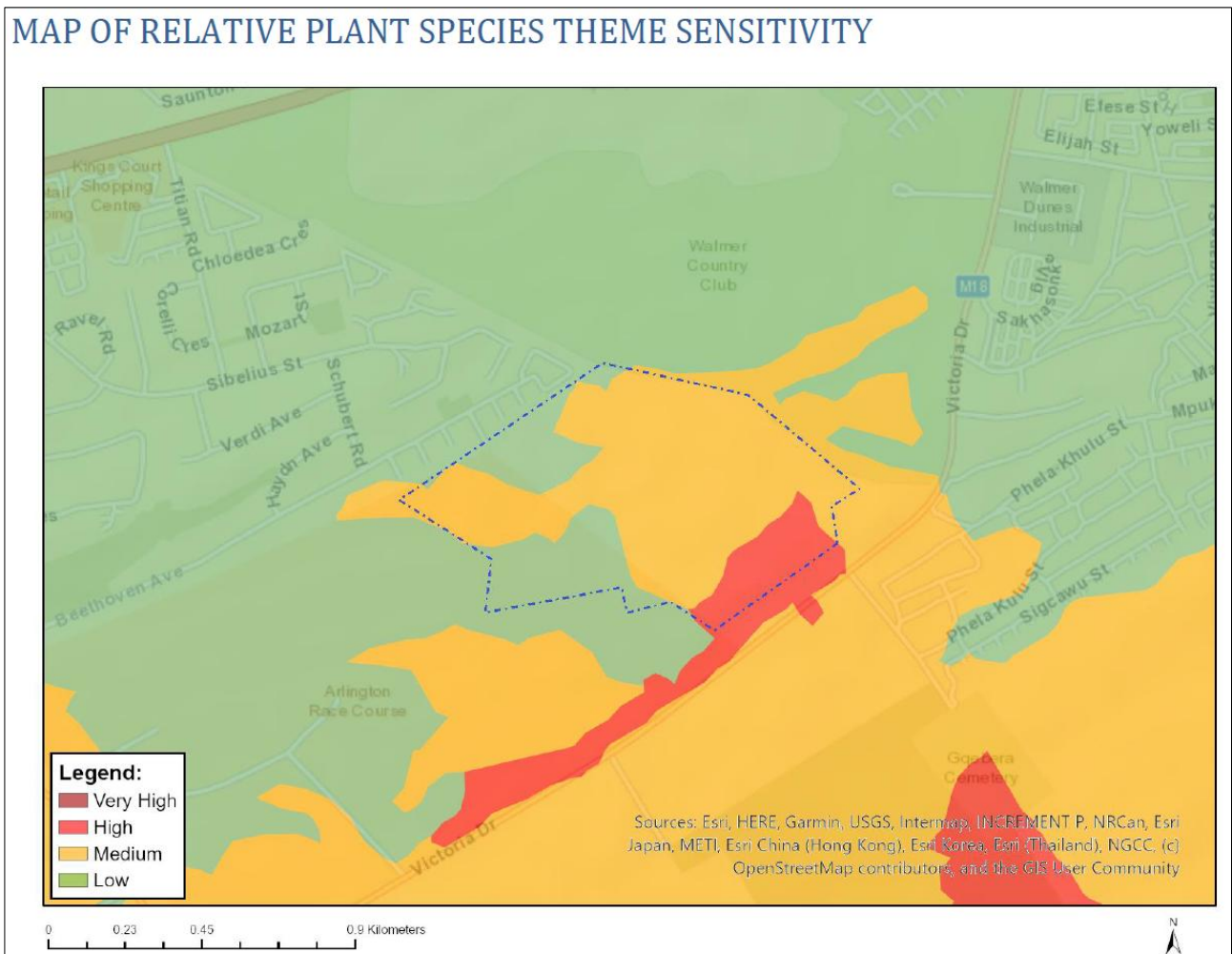


Figure 3.1: Map of the relative plant species theme sensitivity for the Primary PAOI.

Based on the site sensitivity verification (conducted further below) the site sensitivity **is reduced to medium sensitivity**. This is due to the following reasons:

1. None of the plant SCC listed in the Screening Tool Report (see Section 2.1) were identified on site at the time of the site survey.
2. Three plant SCC listed in the Screening Tool Report may plausibly occur on site, as the site hosts patches of indigenous forest/thicket vegetation, which may host these species.
3. No other plant SCC, sensitive or threatened plant species were observed on site.
4. One protected tree was observed on site i.e. *Sideroxylon inerme* subsp. *inerme* (White Milkwood).
5. Six protected plants (protected under the Nature and Environmental Conservation Ordinance (NECO), No. 19 of 1974) were identified on site.
6. The activity will not have an impact on plant SCC beyond the boundary of the Primary PAOI.

Based on the observations, the plant species theme was **assessed as a Compliance Statement** in this report. The terrestrial plant species protocols (GN.R. 1150 of 2020) for a compliance statement (medium sensitivity) were therefore followed for the theme.

Animal Species Theme:

The proposed development area environmental sensitivity for the animal species theme is **identified as high sensitivity** (Figure 3.2).

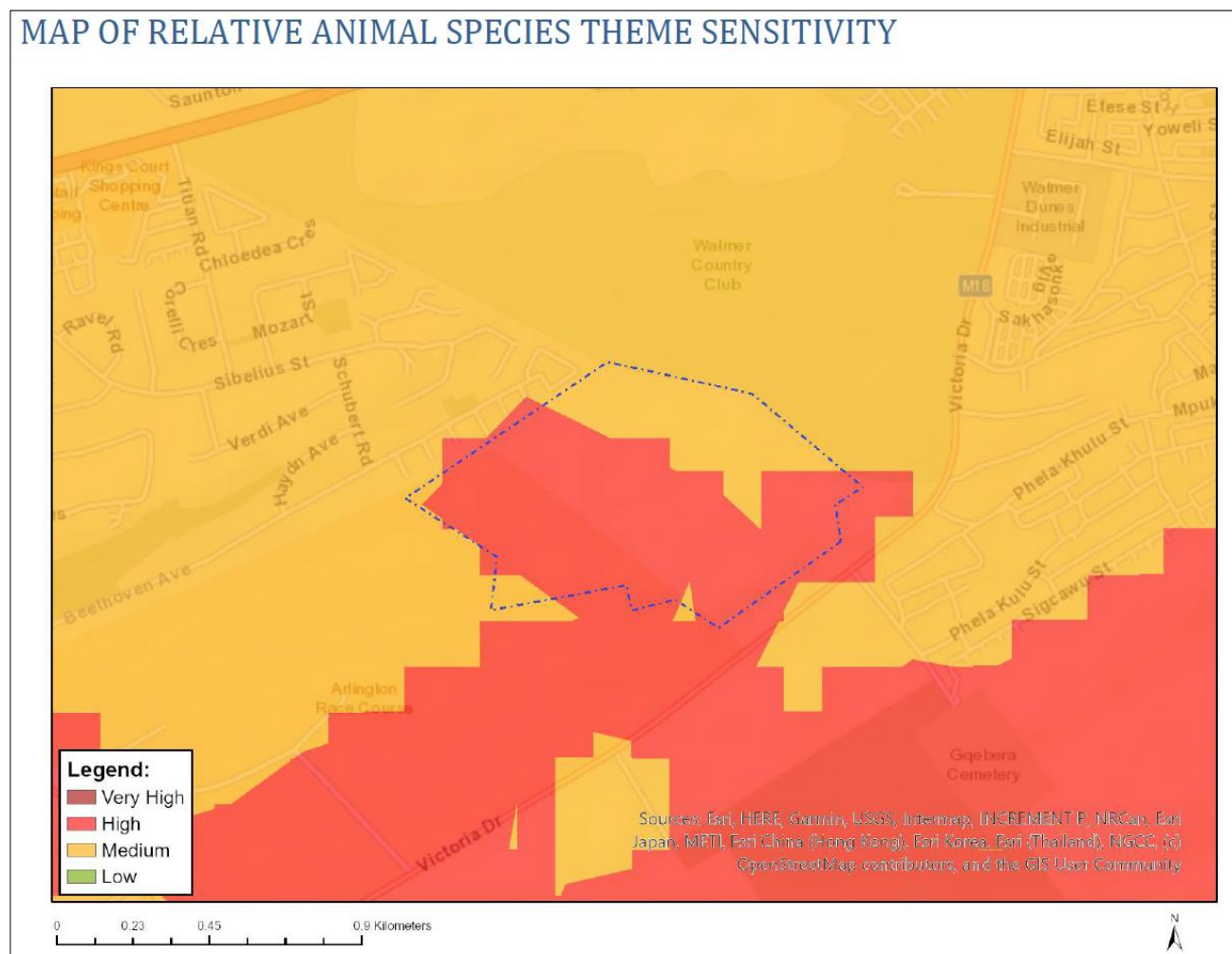


Figure 3.2: Map of the relative animal species theme sensitivity for the Primary PAOI.

Based on the site sensitivity verification (conducted further below) the site sensitivity was **reduced to low sensitivity**. This is due to the following reasons:

1. None of the sensitive animal species listed in the Screening Tool Report (see Section 2.2) were observed on site at the time of the site survey.
2. Nine sensitive animal species are listed in the Screening Tool Report – five bird, two insect and two mammal species.
3. Mostly high-sensitivity bird species were identified in the Screening Tool Report. Given the high degree of past and current disturbance in the greatest portion of the site, as well as in the surrounding areas, these species are unlikely to breed or nest on site and will be able to relocate to the green open space located northeast and southwest of the site.
4. No other sensitive or threatened animal species or tracks/traces thereof were observed on site.
5. The activity will not have an impact on SCC beyond the boundary of the project site/PAOI.

Based on the observations, the animal species theme was **assessed as a Compliance Statement** in this report. The terrestrial animal species protocols (GN.R. 1150 of 2020) for a compliance statement (low sensitivity) were therefore followed for the theme.

Terrestrial Biodiversity Theme:

The proposed development area environmental sensitivity for the terrestrial biodiversity theme is **identified as very high sensitivity** (Figure 3.3).

Based on the site sensitivity verification (conducted further below) the site sensitivity **remains as very high sensitivity**. This is due to the following reasons:

1. The site is not located within nor near any surface or groundwater strategic water source areas (SWSA's).
2. Vegetation on site is largely mapped and described as FFs29 Algoa Sandstone Fynbos (VegMap, 2018/2024), which is classified as a Critically Endangered ecosystem (NEMBA, 2022; NBA, 2025) – on a national level. However, the remnant patches of indigenous vegetation present in the proposed project site are not representative of FFs29 Algoa Sandstone Fynbos and most likely was not in the past, either, before any past vegetation clearing or development was undertaken, or before the site became invaded by Australian Acacia and Eucalyptus trees.
3. The southernmost portions of the site, along the southern boundary, are classified as AT48 Sardinia Forest Thicket (VegMap, 2018/2024). It is, therefore, most likely that vegetation present on site, before settlement of the area, was representative of **AT48 Sardinia Forest Thicket** rather than FFs29 Algoa Sandstone Fynbos. AT48 Sardinia Forest Thicket is also classified as a Critically Endangered ecosystem (NBA, 2025).
4. The municipal-scale vegetation classification and map for the site i.e. the NMBM Final Bioregional Plan (2015), maps the project site as hosting Sardinia Bay Forest Thicket – a Vulnerable ecosystem. The site is, therefore, classed as hosting a Vulnerable vegetation/ecosystem type on a bioregional or municipal scale.
5. The greatest portion of the site has been cleared of the indigenous vegetation originally present on site (for more than the past 22 years) and transformed to grassland dominated by Rooikrans and Port Jackson Willow trees. Only an approx. 15.95 ha area of Sardinia Bay Forest Thicket vegetation remains as separate patches of partially intact vegetation on site – 7.6 ha of which

will not be developed and will be zoned Open Space Zone 3 (conservation usage) (out of the total 43.73 ha project area or Primary PAOI).

6. The site is not classified as a Critical Biodiversity Area (CBA), Ecological Support Area (ESA) or a Critical Ecological Process Area (CEPA) in the NMBM Bioregional Plan (2015).

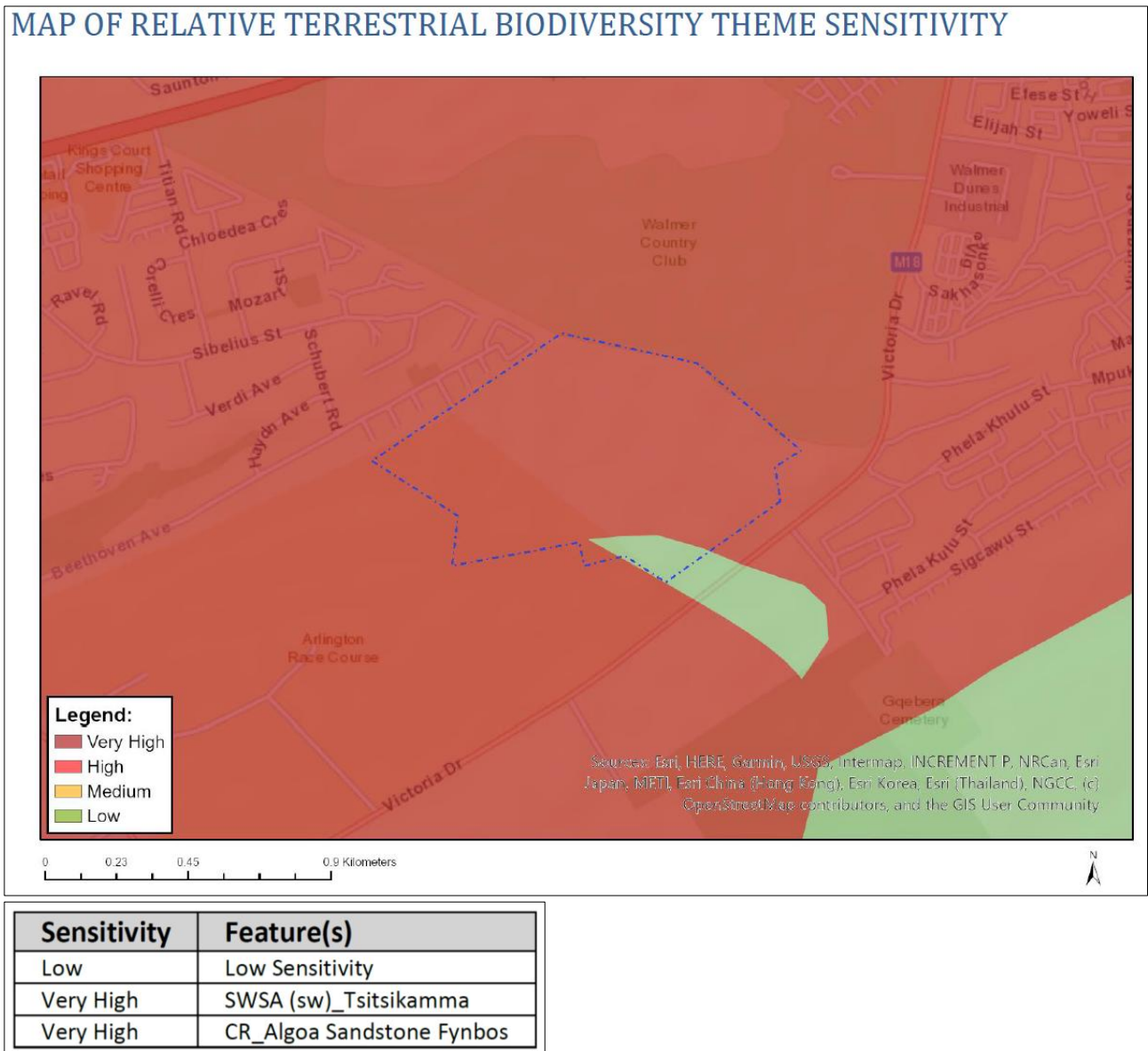


Figure 3.3: Map of the relative terrestrial biodiversity theme sensitivity for the Primary PAOI.

Based on the sensitivity, the terrestrial biodiversity theme was **assessed as a full Terrestrial Biodiversity Impact Assessment** in this report. The terrestrial biodiversity species protocols (GN.R. 1150 of 2020) for a full assessment were therefore followed for the theme.

3.2. Topography

The project site is located between 105 and 115 m above mean sea level. It appears largely flat in topography – however, there are many localized changes in topography due to the presence of uneven ground (due to past cultivation, excavation, mining or dumping of soil, building waste or overburden material from the adjacent developments). There are also small, low-lying, grassy areas

that appear to form ‘pans’ after high rainfall events, with grasses dying back and appearing as bare ground during dry seasons.



Figure 3.4: Topography of the proposed project site (Primary PAOI: outlined in red; Secondary PAOI: outlined in white dashed line) and surrounding areas.

3.3. Local climate

The climate of Port Elizabeth is temperate, subtropical oceanic and is characterized by mild, sunny winters and warm, humid summers. Due to its oceanic climate, the area rarely experiences temperature extremes and frost, and it receives bimodal rainfall i.e. predominantly in autumn and spring. The driest month is June (28 mm) and the wettest month is November (63 mm). February is the hottest month with a mean of 21.6°C, while July is the coldest month with a mean of 13.3°C.

3.4. Geology and soils

The geology of the site is predominantly classed as *Nanaga Formation: Semi-consolidated to consolidated calcareous sandstone and sandy limestone with large-scale crossbedding* – within the *Algoa Group: Marine terrace conglomerate, coquinite, aeolianite, alluvial gravel, sand, silt, calc-tufa, minor gravel*. The Algoa Group consists of six formations which range from Middle Eocene to Late Holocene in age (approx. 41 Ma to 100 Ka). The Nanaga Formation is the third oldest formation within the Algoa Group, underlain by the Bathurst and Alexandria Formations and overlain by the Salnova, Nahoon and Schelm Hoek Formations. A very limited southeastern portion of the site is classed as *Schelm Hoek Formation: Unconsolidated calcareous sand (coasted dunes), minor palaeosols* (Cape Farm Mapper, v. 3).

The area hosts *Greyish, sandy soils* (AR3). Soils are classed as *Imperfectly drained sandy soils, often shallow and often with a plinthic horizon* (Soil ID S4 and S8). These soils may have favourable water-holding properties or they may be seasonally wet – however, they may also be highly erodible due to their sandy nature.

3.5. Land cover and land use

The greatest portion of the site or Primary PAOI consists of green open space i.e. grassland that is invaded by Australian Rooikrans and Port Jackson Willow trees – with a few remnant patches of partially intact forest/thicket vegetation, surrounded by Sweet Thorn (*Vachellia karroo*) trees, present in the northwestern, northern and southeastern portions. The site currently has no specific land use other than green open space or vacant and unused land. The site is used to for the illegal dumping of household waste, garden waste and building rubble; to burn waste and cut firewood, and as a thoroughfare by local residences to travel between suburbs.

According to Google Earth aerial imagery, there is a ruin of an old building or house in the southeastern portion of the site, just west of the reservoir. This building/house was still standing or in use in 2004, after which it was partially demolished. There is also a large soil/overburden mound in the southern portion of the site, northeast of SmartStone Port Elizabeth, with Eucalyptus trees growing on and around it. The reason for its presence is unclear.

An old, dilapidated observation tower of the old Arlington Racecourse is present on the site boundary in the westernmost portion of the site. An area in the central southwestern portion of the site, nearest to the Beaumont Estate, was cultivated until 2018 (at minimum), judging by the presence cultivated blocks separated by hedges and straight furrows in the ground. An area south of the residences in Beaumont Estate, along the central northwestern site boundary, was either cleared of alien invasive trees and levelled, or cleared, excavated and mined for building sand, in the latter half of 2017 (as evidenced by the presence of an area of bare and levelled ground in December 2017). Similarly, a small area of ground in the southeastern portion of the site was cleared of grasses and/or excavated of topsoil at the end of 2010.

The green open space on site is classed as *contiguous low forest & thicket, dense forest & woodland, smallholdings (tree), smallholdings (low veg/grass), smallholdings (bush), urban recreational fields (tree)* and *urban recreational fields (grass)* (see Figure 3.5, below; Cape Farm Mapper, v. 3). Developed areas i.e. residences and businesses/industries, are classed *residential formal (tree), residential formal (low veg/grass), village scattered (bare & low veg/grass combo)* and *industrial*. Very limited areas are classed *low shrubland (fynbos)* and *natural grassland*.

Land uses of the Secondary PAOI include: built-up areas i.e. dirt and tarred roads, low-cost and informal housing, free-standing residences, businesses, estates and townhouse complexes; green open space i.e. grassland with predominantly alien and invasive Acacia trees, a continuous band of forest/thornveld vegetation, an abandoned golf course and horse racecourse with grassland and patches of forest/thornveld; space), and service infrastructure (reservoir, powerlines, telephone lines).

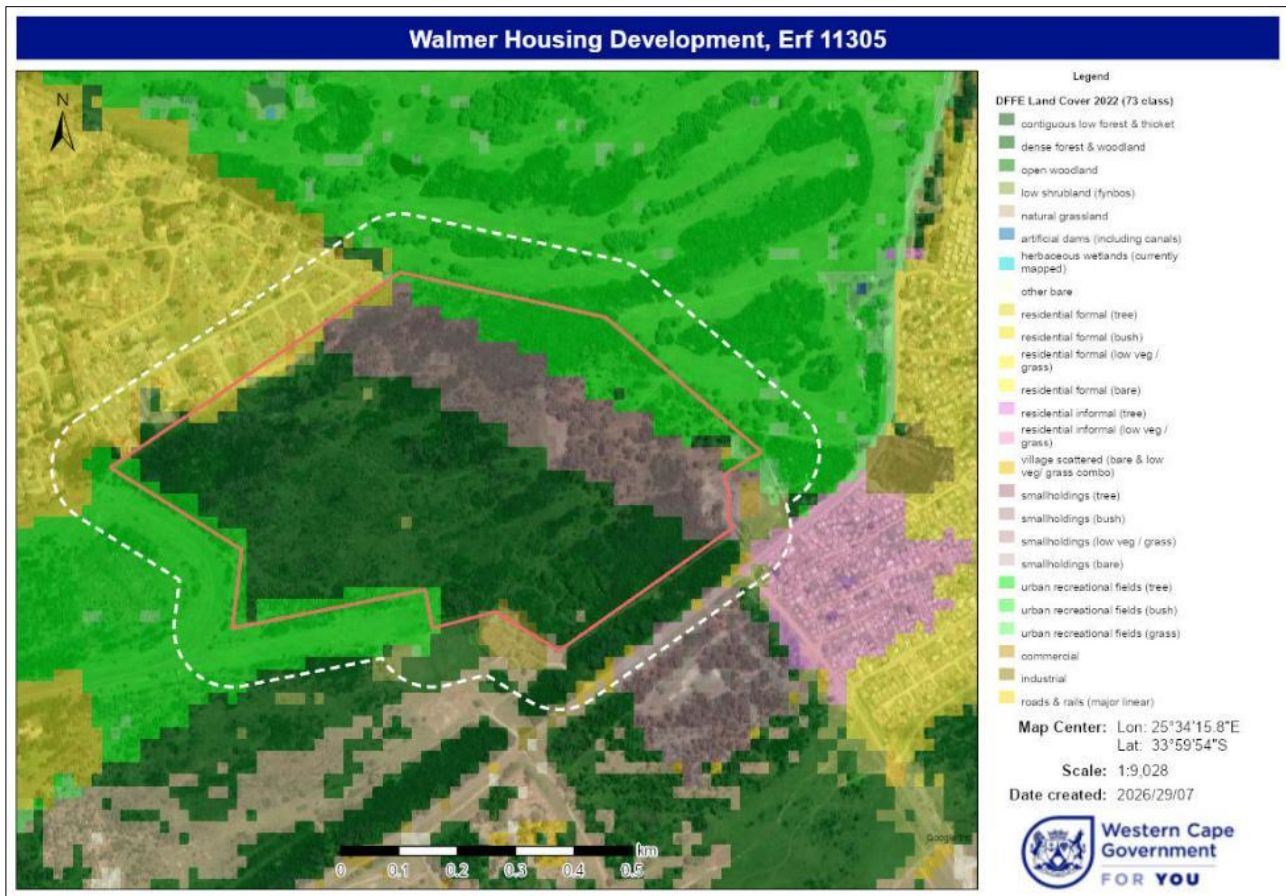


Figure 3.5: Land cover map for the project site and surrounding areas (Primary PAOI: outlined in red; Secondary PAOI: outlined in white dashed line).

Photographs, below, show some of the land uses on site and in the surrounding areas:

Reservoir surrounded by a concrete wall, located east of the site and north of Victoria Drive:



Dense forest/thicket vegetation located on Victoria Drive – southeast of the site and southwest of the reservoir – with a shack and an area that is used to sort and burn waste (presumably from the nearby Arlington Waste Facility):



Dense forest/thicket vegetation on Victoria Drive – southeast of the site:



Area of raised ground/soil or overburden heap with Eucalyptus and Port Jackson Willow trees in the southern/southeastern portion of the site:



Excavated area south of the residences at Beaumont Estate – on the central northern site boundary (facing east):



Facing west on the dirt road that runs along the northern site boundary – residences of Beaumont Estate to right in photograph:



Walking path bordered by grassland, thicket-type shrubs and alien invasive trees in the central northwestern portion of the site:



Dumped garden and household waste surrounded by invaded grassland:



Patches of invasive Castor-oil Plant (*Ricinus communis*) amid grassland, with invasive Port Jackson Willow and indigenous Sweet Thorn trees present along the edge of forest/thicket patches:



Forest/thicket patch in the central portion of site:



Forest/thicket vegetation in the southern portion the site:



Chopped branches of Rooikrans trees amid invaded grassland:



Observation tower on the western site boundary at the old Arlington Racecourse – racecourse to right in photograph:



Forest/thicket patches, Port Jackson Willow and Eucalyptus trees in the southern portion of the site:



3.6. Vegetation

The site was first described according to available literature and databases. A site visit was then conducted to confirm and describe the vegetation as well as the condition of the vegetation unit(s) identified.

3.6.1. Regional-scale vegetation mapping

The South African National Biodiversity Institute (SANBI) vegetation map (called the VegMap, 2024 beta) lists the proposed housing development site predominantly within one vegetation type, namely FFs29 Algoa Sandstone Fynbos (Figure 3.6). It is assigned a threat status of Critically Endangered (NEMBA Revised National List of Ecosystems that are Threatened and in need of Protection (G. NR. 2747, 18 November 2022)).

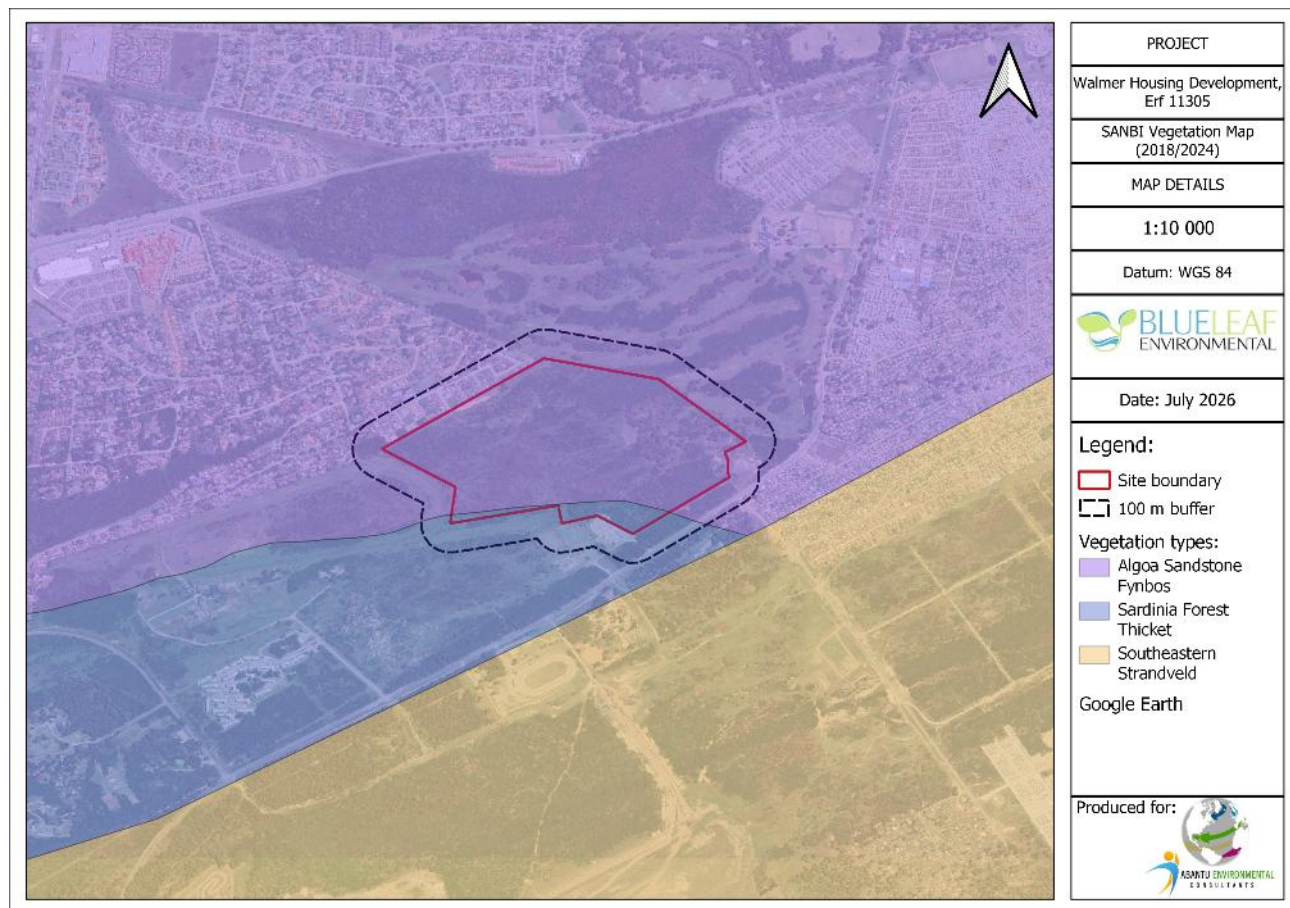


Figure 3.6: SANBI (2018/2024 beta) vegetation map for the site.

However, the patches of indigenous vegetation present in the proposed project site are not representative of FFs29 Algoa Sandstone Fynbos and most likely was not in the past, either, before any past vegetation clearing or development was undertaken, or before the site became invaded by Australian Acacia and Eucalyptus trees. The southernmost portions of the site, along the southern boundary, are classified as AT48 Sardinia Forest Thicket (VegMap, 2018). According to the Red List of Ecosystems, based on the SANBI VegMap (2018) vegetation classifications and published in 2022, AT48 Sardinia Forest Thicket was assigned a threat status of Least Concern (NEMBA Revised National List of Ecosystems that are Threatened and in need of Protection (G. NR. 2747, 18 November 2022)).

The threat status has since been increased to Critically Endangered (NBA, 2025, published on the NBA website: <https://nba.sanbi.org.za>; Harris *et al.*, 2026).

It is most likely that vegetation present on site, before settlement of the area, was representative of AT48 Sardinia Forest Thicket, rather than FFs29 Algoa Sandstone Fynbos.

FFs29 Algoa Sandstone Fynbos

Distribution:

FFs29 Algoa Sandstone Fynbos is only found in the Eastern Cape Province along coastal flats at Port Elizabeth from Van Stadens River in the west to Southdene-Summerstrand in the east, located mostly some kilometres from the coast and close to the coast at only Maitland River Mouth and urbanised Summerstrand. It occurs at altitudes between 20 to 300 m.

Vegetation & Landscape Features:

Algoa Sandstone Fynbos is found on flat to slightly undulating plains supporting grassy shrubland (mainly graminoid fynbos). Grasses become dominant especially in wet habitats. In the south this fynbos unit borders on AT9 Albany Coastal Belt and AZs1 Algoa Dune Strandveld and forms transitional mosaics with both. It also borders on patches of FOz6 Southern Coastal Forest in this area.

Important Taxa:

Tall Shrubs: *Protea eximia*, *P. neriifolia*, *P. repens*. Low Shrubs: *Agathosma hirta*, *A. ovata*, *Erica zeyheriana*, *Euryops ericifolius*, *Helichrysum appendiculatum*, *H. teretifolium*, *Leucadendron salignum*, *L. spissifolium* subsp. *phillipsii*, *Leucospermum cuneiforme*, *Protea cynaroides*, *P. foliosa*, *Tephrosia capensis*. Succulent Herb: *Crassula pellucida* subsp. *marginalis*. Graminoids: *Andropogon eucomus*, *Brachiaria serrata*, *Cymbopogon pospischilii*, *Cynodon dactylon*, *Digitaria eriantha*, *Ehrharta calycina*, *Eustachys paspaloides*, *Ischyrolepis capensis*, *Pentaschistis heptamera*, *P. pallida*, *Thamnochortus cinereus*, *Themeda triandra*, *Tristachya leucothrix*.

Endemic Taxa (^WWetlands):

Low Shrubs: *Agathosma gonaquensis*, *Cyclopia pubescens*^W, *Erica etheliae*.

Conservation:

Conservation target: 23%. About 2% are conserved in the Van Stadens Wild Flower Reserve, The Island Nature Reserve as well as in several private nature reserves. More than 50% transformed (cultivation, urban sprawl of the Nelson Mandela Bay Metropolitan Area). Several Australian *Acacia* species occur as invasive aliens, but only to a limited extent. Erosion moderate and very low.

AT48 Sardinia Forest Thicket

Distribution:

Sardinia Forest Thicket occurs in the Eastern Cape Province – south-east of Port Elizabeth, forming an arc from Seaview in the west to Walmer Heights in the east, and occurring within 2 to 5 km of the coast.

Vegetation & Landscape Features:

Wind and fire-protected areas on moderately undulating coastal dunes. Medium-sized to tall (3 to 5 m) dense thicket, dominated by small trees and woody shrubs, intergrading with small pockets of tree-dominated (*Ekebergia capensis*, *Erythrina caffra*) coastal forest.

Important Taxa (d=dominant, e=South African endemic, et=possibly endemic to a vegetation type):

Tall tree: *Ekebergia capensis* (d), *Erythrina caffra* (d), *Olea capensis*

Small tree: *Pterocelastrus tricuspidatus* (d, e), *Sideroxylon inerme* (d), *Tarchonanthus littoralis* (e), *Vachellia karroo*

Tall shrubs: *Azima tetraacantha* (d), *Olea exasperata* (d, e), *Euclea racemosa* (d), *Searsia glauca* (d, e), *Searsia crenata* (d, e), *Carissa bispinosa*, *Cassine peragua*, *Cussonia thyrsiflora* (e), *Grewia occidentalis* (d), *Gymnosporia buxifolia*, *Gymnosporia capitata* (e), *Maytenus procumbens*, *Mystroxydon aethiopicum* (d), *Robsonodendron maritimum* (e), *Putterlickia pyracantha* (e), *Searsia pterota* (e), *Roepera morgesana*

Succulent tree: *Aloe africana* (d)

Succulent shrub: *Cotyledon velutina* (e)

Geophytic herb: *Veltheimia bracteata* (e)

Graminoid: *Panicum deustum*

Herbaceous climber: *Cynanchum natalitium* (e), *Cynanchum obtusifolium*, *Rhoicissus digitata*, *Solanum africanum* (e)

Woody succulent climber: *Cynanchum viminalis*

Woody climber: *Asparagus aethiopicus*.

Conservation:

Conservation target: 19 %. Protection level: Not protected. Area transformed: 35.99 %. Threats include cultivation, mining and overgrazing. Erosion is low.

Remarks:

(1) The band of grassland or savanna vegetation that occurs between the dune thicket and dune forest community in this thicket type is a consistent feature of the eastern coast of South Africa, right up to the Sand Forests of Maputland. These grassland/savanna communities often contain local endemic herbs and geophytes and probably acted as an important migration route for certain taxonomic groups (e.g. *Chlorophytum*, *Eriosema*, *Tephrosia*, *Trachyandra*) and several coastal grass species. The local unit is the southernmost example of this Thicket-Grassland-Savanna-Coastal Forest association, but it probably extended much further south in previous climatic regimes (Vlok & Euston-Brown 2002).

(2) The thicket species that occur in this unit are shared with AT57 St Francis Dune Thicket.

3.6.2. Municipal-scale vegetation mapping

The NMBM Final Bioregional Plan (2015) was used, as it provides a municipal-scale vegetation map and is the finest-scale bioregional plan available for the NMBM area. The NMBM Final Conservation Assessment and Plan (2010), which was incorporated into the NMBM Final Bioregional Plan (2015), classifies vegetation within the Primary PAOI as predominantly Sardinia Bay Forest Thicket. This vegetation type falls within the Indian Ocean Forest Broad Habitat Unit in the Forest Biome and is assigned an ecosystem threat status of *Vulnerable* (Stewart, 2010).

Sardinia Bay Forest Thicket vegetation as Indian Ocean Forest (typically less than 3.5 m tall) dominated by thicket clumps typical of Algoa Dune Thicket. The matrix is forest characterised by Cape Ash (*Ekebergia capensis*) and Coral Trees (*Erythrina afra*). It is present on aeolian sand (Stewart and Reeves, 2010).

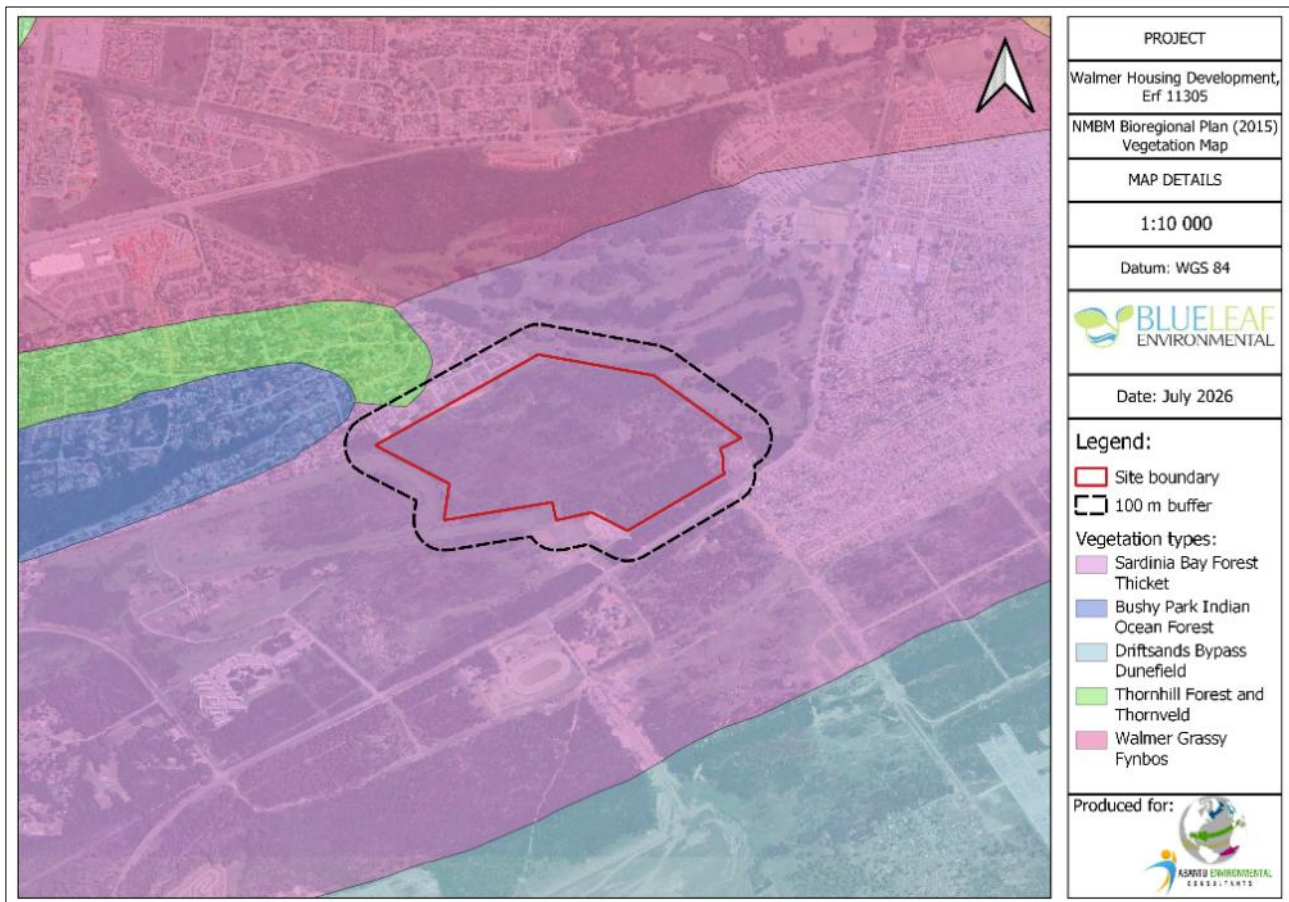


Figure 3.7: NMBM Final Bioregional Plan (2015) vegetation map for the site.

Forest vegetation in the NMBM occurs in a solid form and in a mosaic with thicket and thornveld vegetation. The solid forest vegetation within the NMBM is mostly confined to steep riverine gorges that are sheltered from fire and confined to a band along the wetter southern portion of the municipal area. Threatened forest types within the municipal area include *Critically Endangered* Chelsea Forest Thicket Mosaic, Bushy Park Indian Ocean Forest, Van Stadens Afromontane Indian Ocean Forest, Van Stadens Forest Thicket, Kragga Kamma Indian Ocean Forest and Thornhill Forest and Thornveld (NMBM Draft SDF, 2025).

Vegetation of the Secondary PAOI includes predominantly Sardinia Bay Forest Thicket with a very limited portion, in the northwest, classified as Thornhill Forest and Thornveld. Thornhill Forest and Thornveld vegetation consist of patches of Alexandria Forest with *Afrocarpus falcatus* (Outeniqua Yellowwood) and *Erythrina afra* (Coast Coral Tree) locally common – in grassland with many Sweet Thorn Trees (*Vachellia karroo*) (Stewart and Reeves, 2010). Thornhill Forest and Thornveld vegetation is assigned an ecosystem threat status of *Critically Endangered* (Stewart, 2010).

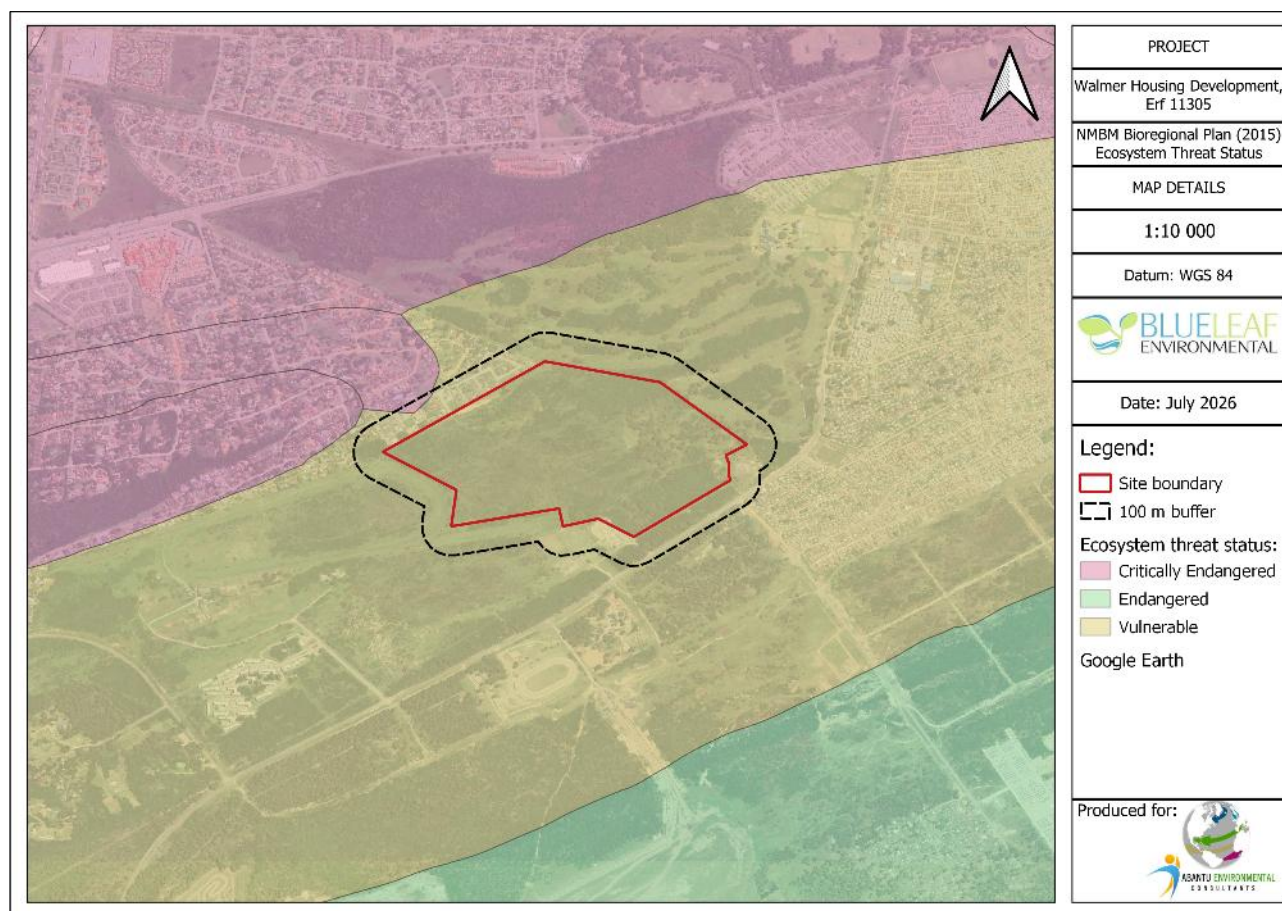


Figure 3.8: NMBM Final Bioregional Plan (2015) ecosystem threat status map for the site.

3.6.3. Fine-scale vegetation mapping

The site visit confirmed the site within the following fine-scale vegetation units (see Figure 3.9). Only vegetation units present within the Primary PAOI are discussed, below.

1. Invaded grassland
2. Forest-thicket patches
3. Built-up or excavated areas.

Existing and old single-track dirt/gravel roads or pathways created over time are indicated as 'roads' on the map.

A patch or small hedge of the alien and invasive species, *Agave sisalana* (Sisal Agave), is present on the southern boundary of the site, with the adjacent SmartStone Port Elizabeth property. This patch of alien plants is indicated on the map but not discussed, below. It is present within 'invaded grassland'. Clumps and individual Eucalyptus trees are also indicated on the map but not discussed, below. Clumps of Eucalyptus trees are predominantly present amid 'forest-thicket patches' – which precludes the growth of forest-thicket type plants in these areas. Smaller clumps and individual Eucalyptus trees are present amid 'invaded grassland' in the central to southeastern portion of the site.

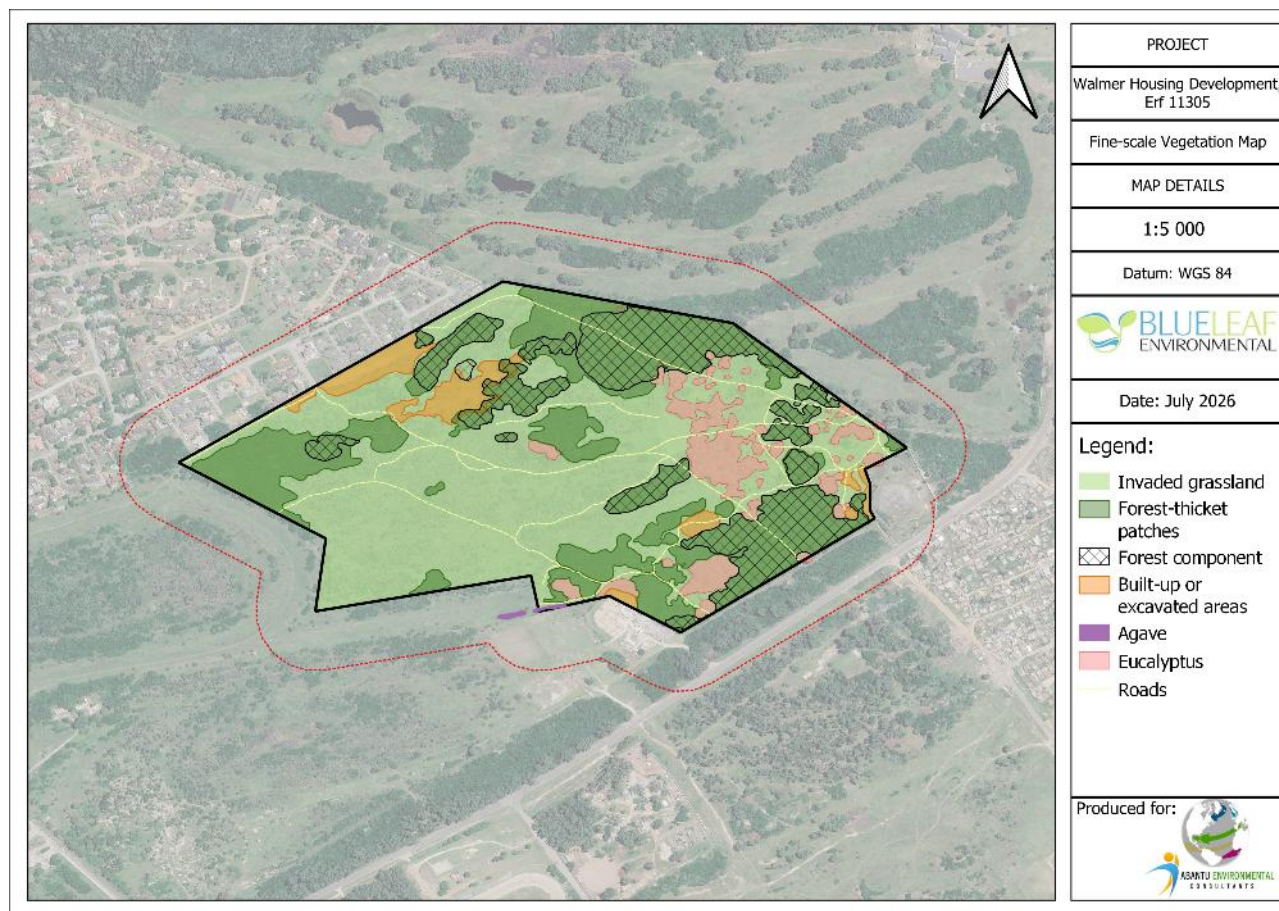


Figure 3.9: Fine-scale vegetation map for the project site.

1. Invaded grassland

The project site consists predominantly of secondary grassland – dominated by Kikuyu, Buffalo and Kweek or Couch Grass – with a moderate to very high density of alien and invasive trees and shrubs present in the northern, western and central portions of the site, in particular. The invaded grassland consists of areas that were cleared of vegetation in the distant past (more than 22 years ago). These areas are no longer representative of the original vegetation type(s) present on site or mapped for the site – presumably Sardinia Forest Thicket/Sardinia Forest Thicket vegetation.

The invaded grassland is dominated by a few grass species i.e. *Pennisetum clandestinum*, *Stenotaphrum secundatum*, *Cynodon dactylon*, *Eragrostis* spp. and *Sporobolus africanus*.

There is a dominant presence of Port Jackson Willow (*Acacia saligna*) and Rooikrans (*Acacia cyclops*) throughout the invaded grassland vegetation unit, as well as localized patches of Castor-oil Plant (*Ricinus communis* var. *communis*), *Euphorbia helioscopia*, *Solanum sisymbriifolium* and *Rapistrum rugosum*. Other alien invasive trees and shrubs present amid grasses, include: *Casuarina* sp. (Beefwood), *Schinus terebinthifolius*, *Cestrum laevigatum*, *Melia azedarach*, *Datura stramonium* and *Lantana camara*.

Indigenous trees, shrubs and forbs present amid grasses, include: *Vachellia karroo*, *Sideroxylon inerme* subsp. *inerme* (White Milkwood), *Searsia* spp., *Scutia myrtina*, *Passerina corymbosa*, *Roepera maritima*, *Helichrysum aureum*, *H. cymosum*, *Senecio inaequidens*, *Hebenstretia*

integrifolia, *Chaenostoma campanulatum* and *Lessertia stenoloba*; bulbs (geophytes), *Brunsvigia gregaria*, *Moraea britteniae*, *Colchicum eucomoides* and *Oxalis* spp., and succulents i.e. *Carpobrotus deliciosus* and *Trachyandra ciliata*.

2. Forest-thicket patches

The remnant patches of forest/thicket vegetation on site are mostly present in the northwestern, northeastern and southeastern portions of the development site. These forest/thicket patches are referred to as 'partially intact' as much of their growth is secondary, particularly around the perimeter of patches i.e. post clearing of vegetation, as evidenced by Google Earth aerial imagery. These patches are also 'partially intact' because there is a significant presence of alien and invasive Eucalyptus, Port Jackson Willow and Rooikrans trees across the development site – in both forest/thicket and invaded grassland vegetation units. Forest/thicket patches cover an approx. 15.95 ha area of the development site i.e. 36% of the total development site.

The 'forest component' of forest-thicket patches/vegetation are also indicated in the fine-scale vegetation map.

It may be assumed that forest/thicket vegetation is representative of the original Sardinia Forest Thicket (VegMap, 2018/2024) or Sardinia Bay Forest Thicket (NMBM Bioregional Plan, 2015) vegetation mapped for the site and adjacent areas. It is dominated by a dense cover of forest and thicket-type trees and shrubs i.e. *Sideroxylon inerme* subsp. *inerme* (White Milkwood), *Vachellia karroo*, *Searsia* spp., *Scutia myrtina*, *Plumbago auriculata*, *Carissa bispinosa*, *Clausena anisata*, *Mystroxydon aethiopicum*, *Schotia afra*, *Putterlickia pyracantha*, *Hypoestes forskoolii* and *Felicia erigeroides*, as well as climbers i.e. *Solanum africanum*, *Cynanchum obtusifolium*, *Clematis brachiata*, *Rhocissus tridentata*, *Zehneria scabra* and *Cyphostemma cirrhosum*.

There is a limited cover of grasses (*Panicum* sp.), geophytes and low-growing shrubs present in between forest/thicket patches.

3. Built-up or excavated areas

Built-up or excavated areas include the old building or house (now a ruin) present in the easternmost portion of the site, just west of the reservoir, as well as areas on site that appear to have been cultivated (presumably to grow vegetables), mined for sand or topsoil, or cleared during construction of surrounding residences, businesses and fences/concrete walls.

Built-up or excavated areas consist of bare ground and secondary vegetation i.e. grasses and alien and invasive shrubs and trees, as well as remnant indigenous trees (e.g. White Milkwood) and thicket-type shrubs.

3.7. Plant species

Appendix A lists all the plant species that were observed on site during the site survey, as well as plant species that may potentially occur on site based on observations made by other users on iNaturalist.org in the adjacent areas, and the vegetation description provided for AT48 Sardinia

Forest Thicket (VegMap, 2018/2024) – i.e. more than 110 plant species, of which 24 are exotic species.

Table 3.1, below, lists all the sensitive species or Species of Conservation Concern (SCC) that may occur on site. This includes all the plant species listed in the Screening Tool Report, regardless of being found on site or not. According to SANBI, the names of some of the species identified in the Screening Tool Report may not appear in the Final BAR/Scoping & EIA Reports nor any of the specialist reports released into the public domain and are therefore just referred to as “Sensitive species #”.

Table 3.1: List of plant SCC that may occur in the study area.

Family	Scientific name	Red List (2024)	Observed (Y/N)	Likelihood of occurrence on site
Aizoaceae	<i>Corpuscularia lehmannii</i>	Critically Endangered	N	Unlikely
Apiaceae	<i>Centella tridentata</i> var. <i>hermanniifolia</i>	Rare	N	Unlikely
Asteraceae	<i>Caputia scaposa</i> var. <i>addoensis</i>	Endangered	N	Unlikely
Celastraceae	<i>Gymnosporia elliptica</i>	Vulnerable	N	Unlikely
Ericaceae	<i>Erica chloroloma</i>	Vulnerable	N	Unlikely
Ericaceae	<i>Erica glumiflora</i>	Vulnerable	N	Unlikely
Ericaceae	<i>Erica zeyheriana</i>	Vulnerable	N	Unlikely
Fabaceae	<i>Argyrolobium crassifolium</i>	Endangered	N	Unlikely
Fabaceae	<i>Aspalathus recurvispina</i>	Critically Endangered	N	Unlikely
Fabaceae	<i>Lotononis acuminata</i>	Vulnerable	N	Unlikely
Myrsinaceae	<i>Rapanea gilliana</i>	Endangered	N	Plausible
Orchidaceae	<i>Holothrix longicornu</i>	Critically Endangered	N	Unlikely
Rutaceae	<i>Agathosma gonaquensis</i>	Critically Endangered	N	Unlikely
Rutaceae	<i>Agathosma stenopetala</i>	Vulnerable	N	Plausible
Scrophulariaceae	<i>Selago rotundifolia</i>	Vulnerable	N	Unlikely
Not provided	Sensitive species 448	Vulnerable	N	Unlikely
Not provided	Sensitive species 588	Vulnerable	N	Unlikely
Not provided	Sensitive species 654	Vulnerable	N	Unlikely
Not provided	Sensitive species 657	Endangered	N	Unlikely
Not provided	Sensitive species 670	Vulnerable	N	Unlikely
Not provided	Sensitive species 991	Critically Endangered	N	Unlikely
Not provided	Sensitive species 1252	Vulnerable	N	Plausible

* As per the best practice guideline that accompanies the protocol and screening tool, the name of these sensitive species may not appear in the Final BAR/Scoping & EIA Reports nor any of the specialist reports released into the public domain.

Table 3.2, below, lists protected plants and trees observed on site, as well as potentially present on site, and which require permits from DEDEAT and DFFE before removal and clearing.

Table 3.2: Protected plants and trees observed in the study area.

Family	Scientific name	Common name	Red List (2024)	PNCO (1974)	NFA (2023)
Aizoaceae	<i>Carpobrotus deliciosus</i> (L.Bolus) L.Bolus	Sweet Sour Fig	LC	X	
Aizoaceae	<i>Mesembryanthemum aitonis</i> Jacq.	Sea Spinach	LC	X	
Amoryllidaceae	<i>Brunsvigia gregaria</i> R.A.Dyer	Albany Candelabra	LC	X	
Asphodelaceae	<i>Aloe africana</i> Mill.	Uitenhage Aloe	LC	X	
Hyacinthaceae	<i>Veltheimia bracteata</i> Harv. ex Baker	Forest Lily	LC	X	

Family	Scientific name	Common name	Red List (2024)	PNCO (1974)	NFA (2023)
Iridaceae	<i>Moraea britteniae</i> (L.Bolus) Goldblatt		LC	X	
Pittosporaceae	<i>Pittosporum viridiflorum</i> Sims	Cheesewood	LC		X
Rutaceae	<i>Clausena anisata</i> (Willd.) Hook.f. ex Benth. var. <i>anisata</i>	Horsewood	LC	X	
Rutaceae	<i>Vepris lanceolata</i> (Lam.) G.Don	White Ironwood	LC	X	
Rutaceae	<i>Zanthoxylum capense</i> (Thunb.) Harv.	Knobwood	LC	X	
Sapotaceae	<i>Sideroxylon inerme</i> L. subsp. <i>inerme</i>	White Milkwood	LC		X

Below are photographs of some of the plant species observed on site:

<i>Trachyandra ciliata</i>	<i>Colchicum eucomoides</i>	<i>Datura stramonium</i> (exotic)
		
<i>Ricinus communis</i> (exotic)	<i>Solanum sisymbriifolium</i> (exotic)	<i>Mystroxydon aethiopicum</i>
		
<i>Cyphostemma cirrhosum</i> (climber) on <i>Vachellia karroo</i>	<i>Sideroxylon inerme</i>	
		

3.8. Animal species

Table 3.3, below, lists all the animal Species of Conservation Concern (SCC) listed in the Screening Tool Report. According to SANBI, the names of some of the species identified in the Screening Tool Report may not appear in the Final BAR/Scoping & EIA Reports nor any of the specialist reports released into the public domain and are therefore just referred to as “Sensitive Species #”.

None of the animal SCC listed in Table 3.3 were observed on site. No other sensitive animal species were observed on site. Species that are likely to occur on site or plausibly visit the site, include: the Knysna Warbler, African Marsh Harrier, Crowned Eagle, Denham's Bustard, the Algoa Opal Butterfly and the Duthie's Golden Mole.

As it was cold and rainy at the time of the site survey, very few animal species would be active and, therefore, very few animals were observed on site during the site visit (other than commonly occurring bird species). Observations of animals made in the general area (by other users on iNaturalist.org), include:

Mammals:

Cape Gray Mongoose, Vervet Monkey, Cape Golden Mole, Hottentot Golden Mole, Cape Mole-rat, Least Dwarf Shrew, Cape Genet, Woodland Dormouse, Southern African Vlei Rat.

Birds:

Knysna Warbler, Cape Weaver, Southern Masked Weaver, Forest Buzzard, Hadedda Ibis, African Harrier-hawk, Greater Double-collared Sunbird, Southern Double-collared Sunbird, Cape Robin-chat, Cape Sparrow, Fiscal Flycatcher, Little Sparrowhawk, Cape White-eye, Cape Bulbul, Collared Sunbird, Western Cattle Egret, Cape Wagtail, Scaly-throated Honeyguide, Black-backed Puffback, Forest Canary, Green-backed Camaroptera, African Paradise Flycatcher, Spotted Eagle-owl, African Dusky Flycatcher, Grey Sunbird, Speckled Mousebird, Black-collared Barbet, Laughing Dove, Red-eyed Dove, Bronze Mannikin, African Crested Flycatcher, Olive Thrush, Knysna Touraco, Sombre Greenbul, Common Waxbill, Terrestrial Brownbul, Black-bellied Starling, Black-headed Oriole, Cape Batis, Speckled Pigeon, Egyptian Goose, Amethyst Sunbird, Olive Woodpecker, Southern Red Bishop, Fork-tailed Drongo, Hamerkop, Brown-hooded Kingfisher, Burchell's Coucal, Little Heron, Rufous-naped Lark, Red-winged Starling, Southern Red-fronted Tinkerbird, Common Starling, Yellow Bishop, Southern Boubou, Pied Crow, Spotted Thick-knee, Black Cuckoo, Southern Fiscal Shrike, Streaky-headed Seedeater, Helmeted Guinea-fowl, Black Sparrowhawk, Red-billed Quelea, Greater Striped Swallow, Rock Dove, Barn Swallow, Yellow-billed Duck.

Amphibians:

Eastern Leopard Toad, Raucous Toad, Painted Reed Frog, Delalande's River Frog, Gray's Stream Frog, Cape River Frog, Bronze Caco.

Reptiles:

Boomslang, Southern Dwarf Chameleon, Puff Adder, Olive Snake, Eastern Cape Legless Skink, Western Natal Green Snake, Spotted Skaapsteker, Red-sided Skink, Leopard Tortoise, Cape Grass Lizard.

Insects:

Butterflies: Blue Pansy, Forest White, Rainforest Brown, Green-banded Swallowtail, Eastern Cape Autumn Widow, Meadow White, African Plain Tiger, African Veined White, Common Zebra Blue, Eastern Pointed Paper White, Southern Round-winged Orange Tip, Common Fig-tree Blue, Painted Lady, Common Geranium Bronze, Cupreous Blue.

Moths: Extorted Tussock, Pine Tree Emperor, Orange Drab, Specious Tiger, Oleander Hawk Moth, Cabbage Tree Emperor, Red-tailed Gypsy Moth, Herald.

Other: Giant Carpenter Bee, Garden Locust, Cape Mountain Cockroach, Paper Wasp, White-spotted Fruit-chaffer, Afrotropical Carpenter Bee, Orange-winged Dropwing Dragonfly, Western Honey Bee.

Table 3.3: List of animal SCC that may occur in the study area.

PHOTOGRAPH	SPECIES NAME	COMMON NAME	SENSITIVITY STATUS	OCCURENCE ON SITE
Birds:				
	<i>Bradypterus sylvaticus</i>	Knysna Warbler	Vulnerable	Likely, as it prefers forest and riverine woodland
	<i>Circus ranivorus</i>	African Marsh Harrier	Endangered	Plausible
	<i>Eupodotis senegalensis</i>	White-bellied Bustard	Vulnerable	Unlikely
	<i>Neotis denhami</i>	Denham's Bustard	Vulnerable	Plausible
	<i>Stephanoaetus coronatus</i>	Crowned Eagle	Near Threatened	Plausible

PHOTOGRAPH	SPECIES NAME	COMMON NAME	SENSITIVITY STATUS	OCCURENCE ON SITE
Invertebrata:				
	<i>Aneuryphymus montanus</i>	Yellow-winged Agile Grasshopper	Vulnerable	Unlikely
	<i>Chrysoritis thysbe whitei</i>	Algoa Opal Butterfly	Vulnerable	Plausible, but unlikely as it prefers coastal shrubland
Mammals:				
	<i>Chlorotalpa duthieae</i>	Duthie's Golden Mole	Vulnerable	Likely
Not provided	Sensitive species 8*	Not provided	Vulnerable	Unlikely

* As per the best practice guideline that accompanies the protocol and screening tool, the name of these sensitive species may not appear in the final BAR report nor any of the specialist reports released into the public domain.

Algoa Opal Butterfly

The Algoa Opal is a rare, range-restricted copper butterfly endemic to the Port Elizabeth area. Recent taxonomic revisions have also elevated this taxon to a distinct species level as *Chrysoritis whitei*.

It occurs in small, fragmented pockets around Port Elizabeth i.e. vegetated coastal duneveld and low scrubby coastal habitats. It is classified as Vulnerable on the SANBI Red List of South African Species due to a tiny extent of occurrence (roughly 69 km²). Threats include urbanization – housing developments and expanding city infrastructure have destroyed historical local subpopulations (such as those previously found in Walmer); - and alien invasive plants. Most remaining known locations survive inside managed public or private local nature reserves.

Knysna Warbler

The Knysna Warbler is extremely secretive, and its presence is normally revealed only during the breeding season when it sings. The Knysna Warbler occurs along the edges of Afro-temperate forests and in thick, tangled vegetation along the banks of watercourses or drainage lines in forest

patches in the Fynbos Biome (Berruti, 1997). It has adapted to thickets of introduced *Rubus* brambles, where it can be locally common, particularly in Western Cape. Birds have recently started to abandon natural, protected forests in favour of narrow belts of suburban riverine woodland with a dense, tangled understorey (Pryke *et al.*, 2011). This was attributed to the availability of nesting substrate as opposed to prey availability (Pryke *et al.*, 2011). In parts of the range, such around George, the species is also regularly recorded in overgrown suburban gardens with dense understorey vegetation (PJ Ginn pers. comm.).

The Knysna Warbler is insectivorous, taking a range of invertebrates, but favouring arachnids and amphipods (Visser and Hockey, 2002). It breeds in spring and early summer, August to December (Smith, 2005), at which time local invertebrates are at their most abundant (Pryke and Samways, 2008).

This species may visit the patches of dense forest/thicket vegetation present on site. This vegetation will largely not be impacted by the proposed development.

African Marsh Harrier

This species is dependent on permanent wetlands, both inland and coastal, for breeding, feeding and roosting. It also hunts over drier floodplains, grasslands, croplands, and Fynbos, where it preys mainly on small rodents (Simmons, 2005), as well as birds, reptiles, frogs and insects. Nests are usually placed in reedbeds over water, although breeding has been recorded in adjacent sedges, Fynbos, scrub and agricultural fields, but these are rare occurrences (Kemp and Kemp, 2006). Breeding adults are largely sedentary (Simmons, 2005) with pairs often retaining the same territory year after year (Simmons, 1990), while juveniles disperse widely. The generation length of 7.8 years was calculated from the mean of two calculated values derived from published and/or extrapolated estimates of mean age at first breeding, maximum longevity in the wild and mean annual adult survival (BirdLife International, 2014).

This species may visit the dams, wetlands and open grassland present in surrounding areas to feed but it is unlikely to roost on site given the high degree of disturbance (development of roads, businesses and residences in surrounding areas).

Denham's Bustard

Denham's Bustard is usually encountered alone or in small groups of c. 2 to 7 birds although occasional aggregations as large as 28 individuals have been reported (Allan, 2002). Displaying male Denham's Bustards were recorded in the Overberg region of Western Cape between August and December with egg-laying taking place in September to October (Allan, 2002). Unfledged young were recorded between September and January. Group sizes in this area were significantly smaller in the breeding season (September to January) than in the non-breeding season (Allan, 2002). Both sexes tend to form single-sex groups when flocking (Allan, 2002). Groups containing a single male and female are rare, indicating a possible polygamous mating system (Allan, 2002) as is usual for bustards (Johnsgard and Jones, 1991). A generation length of 10.3 years is provided by BirdLife International (2014). The species is omnivorous.

This species may visit open grassland areas to rest and forage, but it is unlikely to breed and nest on site.

Crowned Eagle

The species is found mostly in forest, including gallery and riverine forest, but also occurs in woodland and forested gorges in savannah and grassland (Simmons, 2005). Crowned Eagles are readily found in plantations of exotic trees. They normally perch for long periods, resting inside the forest canopy, but will sometimes soar high above the canopy. Males also perform repeated aerial displays accompanied by loud calls in the vicinity of the nesting territory. The diet is composed primarily of mammals (96%) and particularly hyraxes, antelope and primates; large birds (3.5%) and reptiles (0.5%) make up the remainder (Simmons, 2005). A generation length of 18.5 years, based on a published estimate of mean age at first breeding and extrapolated maximum longevity in the wild, was provided by BirdLife International (2014).

This species may visit the patches of dense forest/thicket vegetation present on site. This vegetation will largely not be impacted by the proposed development. It is unlikely to roost on site given the high degree of disturbance (development of roads, businesses and residences in surrounding areas).

Duthie's Golden Mole

Occurs on alluvial sands and sandy loams in Southern Cape Afrotropical forests (especially coastal platform and scarp forest patches) in the Fynbos and Moist Savanna biomes. Coexists with *Amblysomus corriae* in parts of their range, but trapping data suggest that *A. c. corriae* prefers fynbos and forest fringes, and *C. duthieae* deeper forest. Thrives in cultivated areas and gardens. Adults are solitary, but up to four individuals per hectare have been trapped on the same night, suggesting that population densities are relatively high in areas of suitable habitat. Construct shallow subsurface foraging tunnels that radiate outwards from under the roots of trees. Active mainly at night.

This species may plausibly occur on site.

Yellow-winged Agile Grasshopper

The species is associated with fynbos vegetation, where it has been collected "amongst partly burnt stands of evergreen Sclerophyll in rocky foothills" (Brown, 1960). It prefers south-facing cool slopes (Kinvig, 2005).

This species is highly unlikely to occur on site, as it has only been observed in the Western Cape.

Sensitive species 8

This is a reclusive species that is highly unlikely to occur on site as the site consists predominantly of previously-cleared areas and relatively open vegetation – whereas this species prefers dense forest and woodland vegetation.

3.9. Terrestrial Biodiversity

This report makes use of various spatial databases to define the terrestrial ecology of the area associated with the proposed housing development. These include (but are not limited to):

1. Nelson Mandela Bay Municipality Bioregional Plan
2. Strategic Water Source Areas
3. National Freshwater Ecosystem Priority Areas

4. NEMBA Threatened Ecosystems
5. Protected areas.

3.9.1. Nelson Mandela Bay Municipality Bioregional Plan

The NMBM Final Bioregional Plan (Gazette No. 3362, gazetted 30 March 2015) is the only fine-scale bioregional plan available for the Nelson Mandela Bay Municipality. It is a spatial plan that identifies terrestrial and aquatic features essential for conserving biodiversity and maintaining ecosystem functioning. These features are captured and mapped as Critical Biodiversity Areas, Ecological Support Areas and Critical Ecological Process Areas. CBA's include Critically Endangered and Endangered vegetation or ecosystem types and forest patches.

The project site is not classified as a CBA or ESA (Figure 3.10). The project site is not included within a Critical Ecosystem Process area (Figure 3.11).

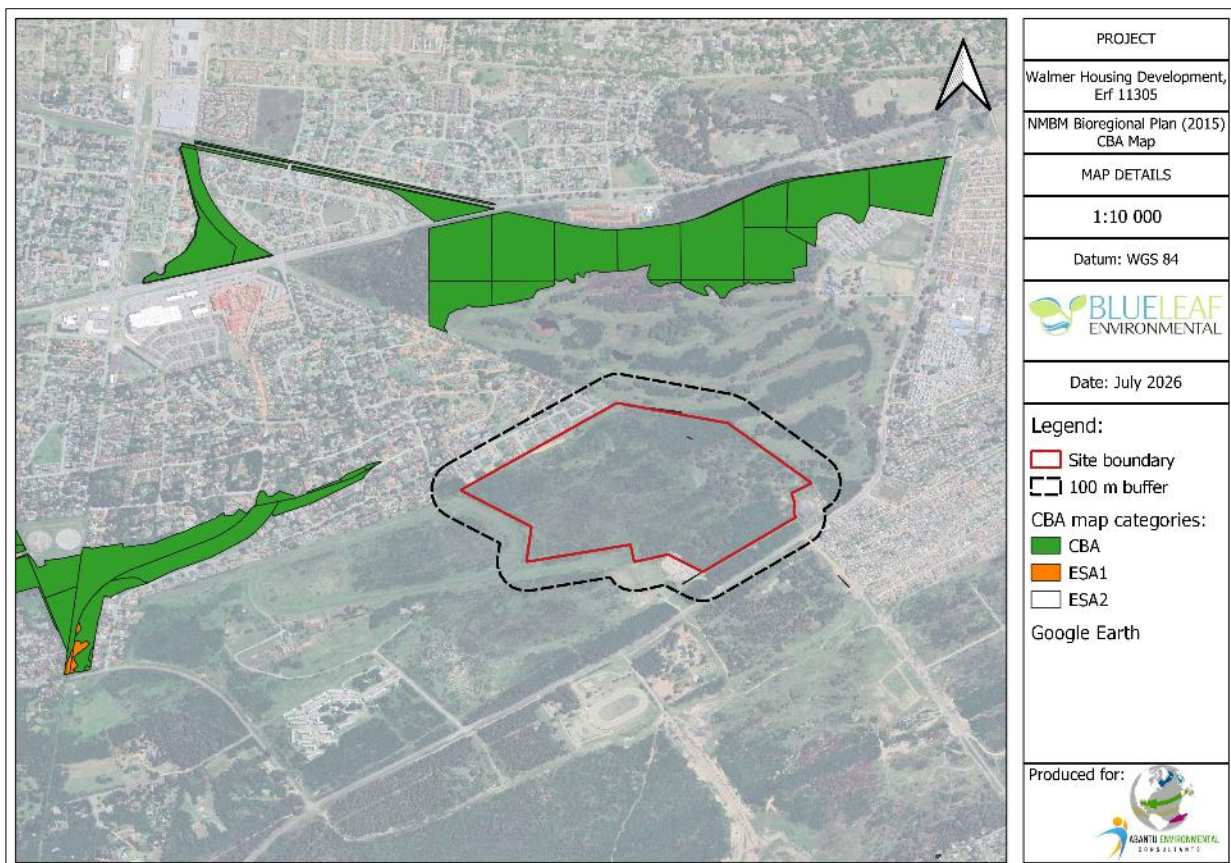


Figure 3.10: NMBM Bioregional Plan CBA map for the project site.

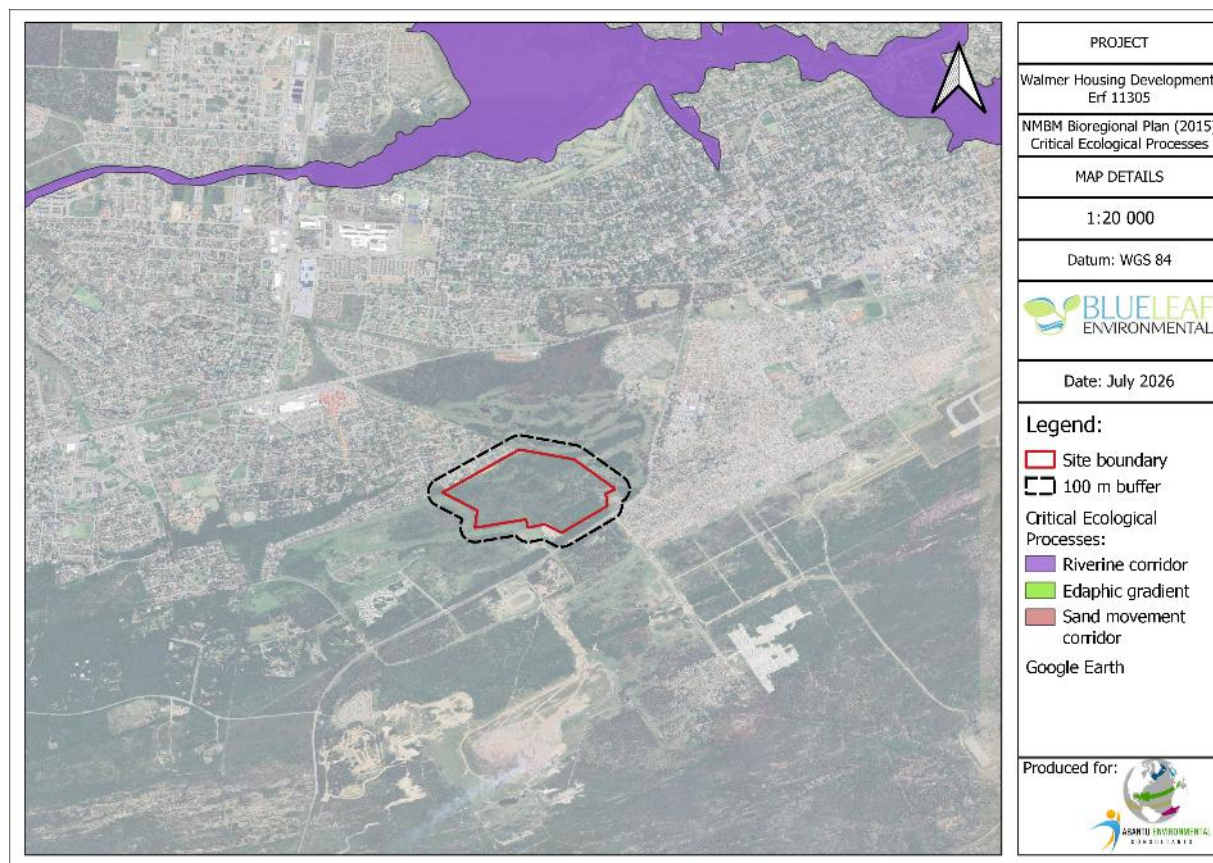


Figure 3.11: NMBM Bioregional Plan Critical Ecological Processes map for the project site.

3.9.2. Strategic Water Source Areas

Strategic Water Source Areas or SWSA's are national ecological infrastructure assets that are essential for water security. These areas of high rainfall make up just 10% of the land area of South Africa, Lesotho and Eswatini, but supply 50% of water to these countries. What happens within the boundaries of SWSAs has an impact on water quality and quantity for millions of people and for economic and agricultural activity downstream. A total of 20 SWSA's were identified. SWSA can also be separated in surface water sources (SWSAsw) and groundwater sources (SWSAgw). Groundwater SWASS are areas which have a high groundwater recharge/availability and are classified as a nationally important resource.

The project site is not located within any surface or groundwater SWSA's.

3.9.3. National Freshwater Ecosystem Priority Areas

The project site is situated in the Algoa sub-water management area (ID 84) in the greater Fish to Tsitsikamma water management area (ID 16). The property is located within the catchment area of the Baakens River (River type: 20_P_U; Present Ecological State (1999): Class C: Moderately modified; River condition: C).

The northernmost portion of the site is classified as a Fish Sanctuary and Associated Sub-quaternary Catchment (Fish FSA) (Nr. 9104; NFEPA, 2011) which is also demarcated with a red fish symbol. Fish sanctuaries are rivers that are essential for protecting threatened and near-threatened freshwater

fish that are indigenous to South Africa. A red fish symbol indicates that there is at least one population of a critically endangered or endangered fish species present within that sub-quaternary catchment (Driver *et al.*, 2011).

The proposed housing development will not have an impact on the Baakens River or the larger sub-quaternary catchment, provided that adequate stormwater management structures are put in place.

3.9.4. Threatened ecosystems

The project site is largely classified as a hosting FFs29 Algoa Sandstone Fynbos – in the Eastern Fynbos-Renosterveld Bioregion – which is listed as a *Critically Endangered* [B1(i)] ecosystem. This is the threat classification for the site according to the NEMBA Revised National List of Ecosystems that are Threatened and in need of Protection (G. NR. 2747, 18 November 2022), which is the latest gazetted list of threatened ecosystems available for the country. It is based on the VegMap (2018) vegetation classification, published under Dayaram *et al.* (2019).

However, the latest vegetation classification available for the region is the VegMap (2024), available as a dataset on the SANBI website (sanbi.org.) and which has not yet been published or gazetted. The National Biodiversity Assessment (2025) published on the NBA website (<https://nba.sanbi.org.za>; Harris *et al.*, 2026) is based on this updated VegMap (2024) vegetation classification. It also classifies vegetation on site as predominantly FFs29 Algoa Sandstone Fynbos, listed as *Critically Endangered* [B1(i)]. However, limited portions along the southern site boundary are classified as AT48 Sardina Forest Thicket (VegMap, 2018). It is also assigned a threat status of *Critically Endangered* (National Biodiversity Assessment (2025) published on the NBA website (<https://nba.sanbi.org.za>; Harris *et al.*, 2026).

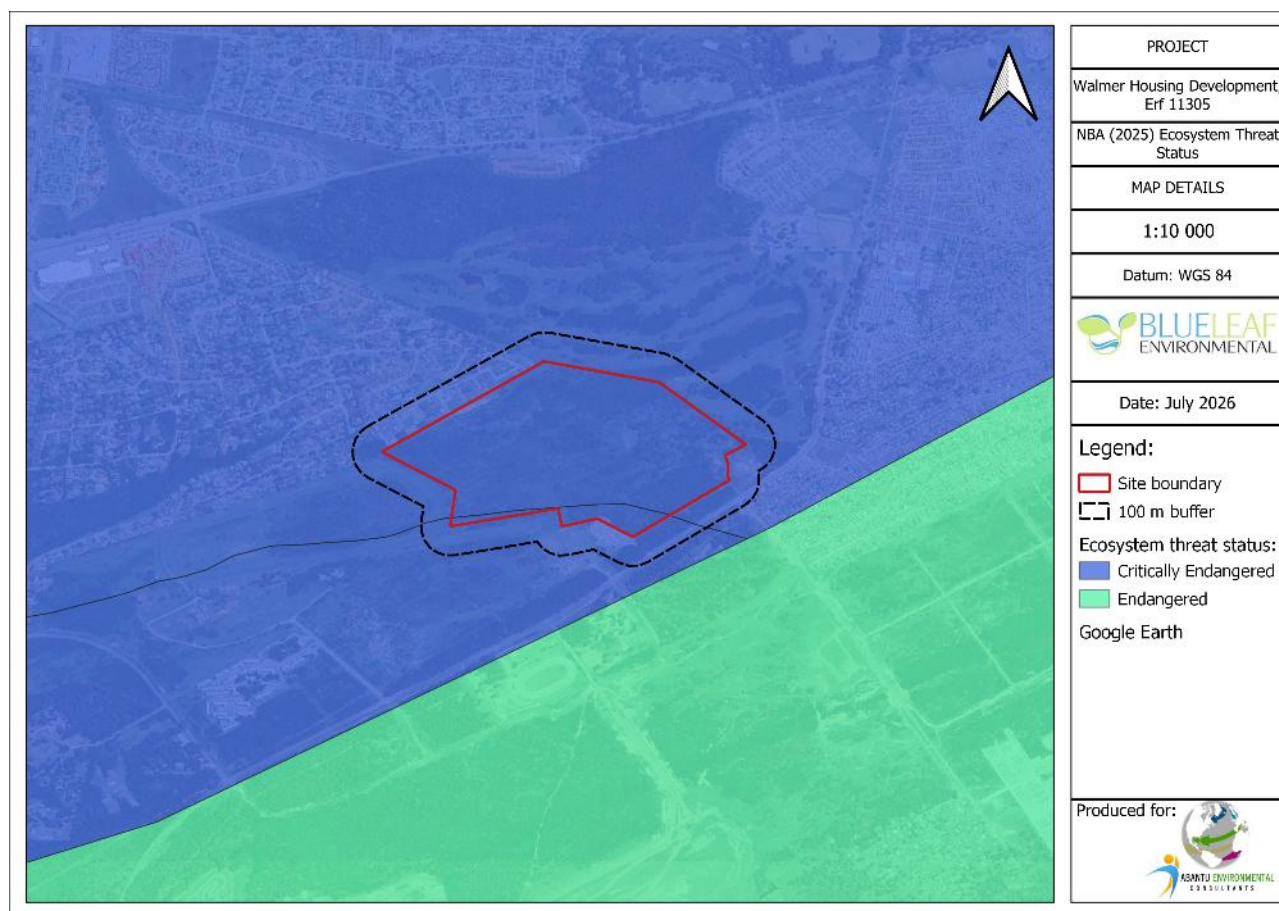


Figure 3.12: National Biodiversity Assessment (2025): Terrestrial Red List of Ecosystems – ecosystem threat status for the site.

3.9.5. Protected areas

The project site is not located within nor near any protected areas or nature reserves. The following protected areas/nature reserve are located within 10 km of the Primary PAOI. These protected areas will not be impacted by the proposed development:

Name of protected area	Distance from project site	Direction from project site
Settlers Park Local Authority Nature Reserve	3.2 km	northeast
Sardinia Bay & Sylvic Local Authority Nature Reserve	4.5 km & 7.9 km	southwest
Sardinia Bay Marine Protected Area	5.5 km	southwest
NMMU Private Nature Reserve	5.6 km	southeast
Cape Recife Local Authority Nature Reserve	7.9 km	southeast
Theescombe Wild Flower Reserve	7.9 km	northwest
Kragga Kamma Game Park	8.8 km	northwest

3.9.6. Nelson Mandela Bay Municipality Draft Spatial Development Framework

The project site is located within the Proposed Urban Edge and Urban Growth Boundary (2015; Figure 3.13).

The site is identified as 'Special Development Incentives Areas: Growth Areas' in the Overall Spatial Development Framework/Spatial Structuring Elements Plan (Spatial Targeted Area 3 – Map 72) (Figure 3.14; NMBM Draft Spatial Development Framework, August 2025).



Figure 3.13: Extract from the Urban Edge Plan showing the locality of the project site (at the yellow square approx.; NMBM Draft SDF, 2025).

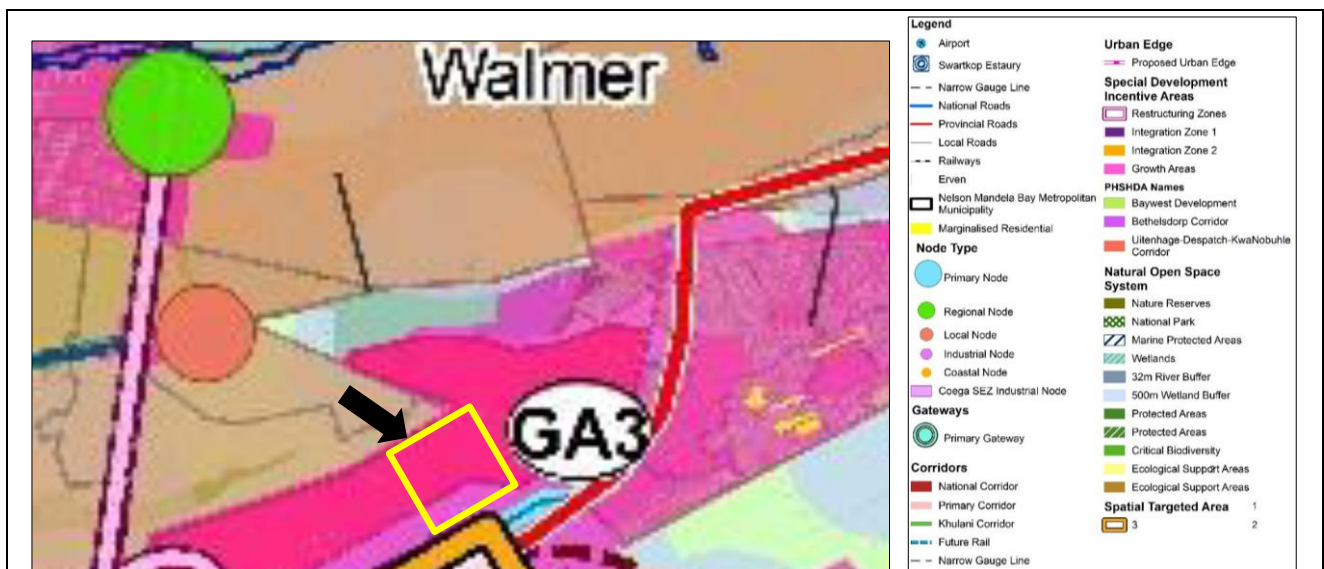


Figure 3.14: Extract from the Overall Spatial Development Framework Plan: Spatial Targeted Area 3, showing the locality of the project site (at the yellow square approx.; NMBM Draft SDF, 2025).

3.10. Exotic and Alien and Invasive Species

It should be noted that the CARA Regulations for the legal obligations regarding alien invasive plants in South Africa have been superseded by the National Environmental Management: Biodiversity Act, 2004 (Act No. 10 of 2004) – Alien and Invasive Species (AIS) Regulations which was promulgated on

1 October 2014. However, CARA has not been repealed and is still included as a reference point to use in terms of the management of AIS within agricultural land.

The National Environmental Management: Biodiversity Act, 2004 (Act No. 10 of 2004) also provides a list of AIS for management (GN. R. 898 of 2014 – the latest amendment published in 2020). AIS are classified into the following categories:

- **Category 1a:** species which must be combatted or eradicated.
- **Category 1b:** species which must be controlled.
- **Category 2:** species which require a permit to carry out a restricted activity within an area specified in the Notice or an area specified in the permit.
- **Category 3:** species which are subject to exemptions.

The following (24) exotic and AIS plants were observed within the project site:

Family	Scientific name	Common name	AIS Category
Amaranthaceae	<i>Chenopodium album</i> L.	White Goosefoot	
Amaranthaceae	<i>Chenopodium murale</i> L. var. <i>murale</i>	Nettle-leaved Goosefoot	
Anacardiaceae	<i>Schinus terebinthifolius</i> Raddi	Brazilian Pepper	Cat 1b
Asparagaceae	<i>Agave sisalana</i> Perrine	Sisal Agave	Cat 2
Asteraceae	<i>Erigeron</i> sp.	Fleabane	
Asteraceae	<i>Hypochaeris radicata</i> L.	Cat's Ear	
Brassicaceae	<i>Rapistrum rugosum</i> (L.) All.	Annual Bastard Cabbage	
Casuarinaceae	<i>Casuarina</i> sp.	Beefwood	Cat 2
Euphorbiaceae	<i>Euphorbia helioscopia</i> L.	Sun Spurge	
Euphorbiaceae	<i>Ricinus communis</i> L. var. <i>communis</i>	Castor-oil Plant	Cat 2
Fabaceae	<i>Acacia cyclops</i> A.Cunn. ex G.Don	Rooikrans	Cat 1b
Fabaceae	<i>Acacia saligna</i> (Labill.) H.L.Wendl.	Port Jackson Willow	Cat 1b
Fabaceae	<i>Melilotus indicus</i> (L.) All.	Annual Yellow Sweetclover	
Fabaceae	<i>Vicia</i> sp.	Vetch	
Fumariaceae	<i>Fumaria muralis</i> Sond. ex W.D.J.Koch subsp. <i>muralis</i>	Wall Furmatory	
Malvaceae	<i>Malva parviflora</i> L. var. <i>parviflora</i>	Cheeseweed	
Meliaceae	<i>Melia azedarach</i> L.	Chinaberry Tree	Cat 1b
Myrtaceae	<i>Eucalyptus</i> sp.	Gum	Cat 2
Poaceae	<i>Pennisetum clandestinum</i> Hochst. ex Chiov.	Kikuyu Grass	
Solanaceae	<i>Cestrum laevigatum</i> Schlttdl.	Inkberry	Cat 1b
Solanaceae	<i>Datura stramonium</i> L.	Jimsonweed	Cat 1b
Solanaceae	<i>Solanum sisymbriifolium</i> Lam.	Sticky Nightshade	Cat 1b
Solanaceae	<i>Physalis peruviana</i> L.	Cape Gooseberry	
Verbenaceae	<i>Lantana camara</i> L.	Lantana	Cat 1b

Exotic species with no AIS classification are not considered as problem species and they do not require specific management measures. They should however be monitored and managed to ensure that they do not become problematic.

4. Site Ecological Importance

The Site Ecological Importance (SEI) rating is determined separately for each environmental theme.

As none of the sensitive animal species or SCC listed in the Screening Tool Report, nor any additional animal SCC were observed on site, and as such species have a low likelihood of occurring on site on a frequent or permanent basis, or breed/nest/roost on site – the Terrestrial Animal Species Theme was reduced from *high* to *low sensitivity* during the site sensitivity verification process (Section 3.1). The Terrestrial Animal Species Theme was, therefore, assessed as a Compliance Statement.

The Terrestrial Plant Species Theme was reduced from *high* to *medium sensitivity* during the site sensitivity verification process, as none of the plant SCC listed in the Screening Tool Report (see Section 3.1) were identified on site at the time of the site survey, and no additional plant SCC, sensitive or threatened plant species were observed on site. Therefore, the Terrestrial Plant Species Theme was assessed as a Terrestrial Plant Species Compliance Statement.

The specialist theme protocols do not require a SEI rating for medium and low sensitivity themes and therefore a SEI rating was not calculated for the Terrestrial Animal and Plant Species Themes in this report.

The Terrestrial Biodiversity Theme was maintained as *very high sensitivity* during the site sensitivity verification process. This is due to the classification of the site as hosting a Critically Endangered ecosystem on a national level (FFs29 Algoa Sandstone Fynbos: NEMBA Revised National List of Ecosystems that are Threatened and in need of Protection, 2022; AT48 Sardinia Forest Thicket: NBA, 2025; Harris *et al.*, 2026) – and a Vulnerable ecosystem on a municipal or bioregional level (Sardinia Bay Forest Thicket: NMBM Bioregional Plan, 2015).

The Terrestrial Biodiversity site sensitivity was determined for the Primary PAOI. Sensitivity was firstly based on the DFFE Screening Tool Report outcome and secondly, a subsequent confirmation of these sensitivities during the site visit. Additional biodiversity related constraints (if any) were also identified and aligned with specific characteristics of the site.

The terrestrial biodiversity theme Site Ecological Importance sensitivity ratings for the fine-scale vegetation units were determined using the methodology prescribed by the Species Environmental Assessment Guideline (SANBI, 2022) and summarised in Section 2.6 of this report. The site sensitivity rating was determined for the Primary PAOI (43.73 ha in extent), as the loss of habitat would be restricted to the direct development footprint or Primary PAOI.

Environmental constraints were identified and aligned with specific characteristics of the site. The following site characteristics contributed to determining an SEI or sensitivity rating for the terrestrial biodiversity present on site:

Components assessed	Sensitivity allocation	Justification
Conservation Importance (CI)	Forest component (of forest-thicket vegetation)	Confirmed or highly likely occurrence of CR, combined with Critically Rare species that have a global EOO of < 10 km ² .

Components assessed	Sensitivity allocation	Justification
		Natural forests patched occur within some of the thickets.
	Forest-thicket patches	Presence of some rare species. Natural habitat of EN ecosystem type.
	Invaded grassland	No confirmed or highly likely populations of SCC.
	Built-up or excavated areas	No confirmed or highly likely populations of range-restricted species. < 50% of receptor contains natural habitat with limited potential to support SCC.
Functional Integrity (FI)	Forest component	High habitat connectivity serving as functional ecological corridors. Minimal current negative ecological impacts with no signs of major past disturbance.
	Forest-thicket patches	Good habitat connectivity with potentially functional ecological corridors and a regularly used road network between intact habitat patches. Only minor current negative ecological impacts (e.g. few livestock utilising area) with no signs of major past disturbance (e.g. ploughing) and good rehabilitation potential.
	Invaded grassland	Almost no habitat connectivity but migrations still possible across some modified or degraded natural habitat and a very busy used road network surrounds the area. Low rehabilitation potential.
	Built-up or excavated areas	Several minor and major current negative ecological impacts.
Biodiversity Importance (BI)	Forest component	Very high
	Forest-thicket	High
	Invaded grassland	Low
	Built-up or excavated areas	Low
Receptor Resilience (RR)	Forest component	Very high Habitat that can recover rapidly (~ less than 5 years) to restore > 75% of the original species composition and functionality of the receptor functionality, or species that have a very high likelihood of remaining at a site even when a disturbance or impact is occurring, or species that have a very high likelihood of returning to a site once the disturbance or impact has been removed.
	Forest-thicket patches	High Habitat that can recover relatively quickly (~ 5–10 years) to restore > 75% of the original species composition and functionality of the receptor functionality, or species that have a high likelihood of remaining at a site even when a disturbance or impact is occurring, or species that have a high likelihood of returning to a site once the disturbance or impact has been removed.
	Invaded grassland	Low

Components assessed	Sensitivity allocation	Justification
	Built-up or excavated areas	Habitat that is unlikely to be able to recover fully after a relatively long period: > 15 years required to restore ~ less than 50% of the original species composition and functionality of the receptor functionality, or species that have a low likelihood of remaining at a site even when a disturbance or impact is occurring, or species that have a low likelihood of returning to a site once the disturbance or impact has been removed.
Site Ecological Importance (SEI) (BI x RR)	Very high for forest component (of forest-thicket vegetation)	Avoidance mitigation – no destructive development activities should be considered. Offset mitigation not acceptable/not possible (i.e. last remaining populations of species, last remaining good condition patches of ecosystems/unique species assemblages). Destructive impacts for species/ecosystems where persistence target remains.
	High for forest-thicket patches	Avoidance mitigation wherever possible. Minimisation mitigation – changes to project infrastructure design to limit the amount of habitat impacted; limited development activities of low impact acceptable. Offset mitigation are not required.
	Low for invaded grassland	Minimisation and restoration mitigation – development activities of medium to high impact acceptable followed by appropriate restoration activities.
	Low for built-up or excavated areas	Minimisation and restoration mitigation – development activities of medium to high impact acceptable followed by appropriate restoration activities.

A SEI sensitivity map was developed for the terrestrial biodiversity theme assessed in this report (Figure 4.1). This map was developed based on the identified environmental characteristics found within the Primary PAOI and listed in the Table, above.

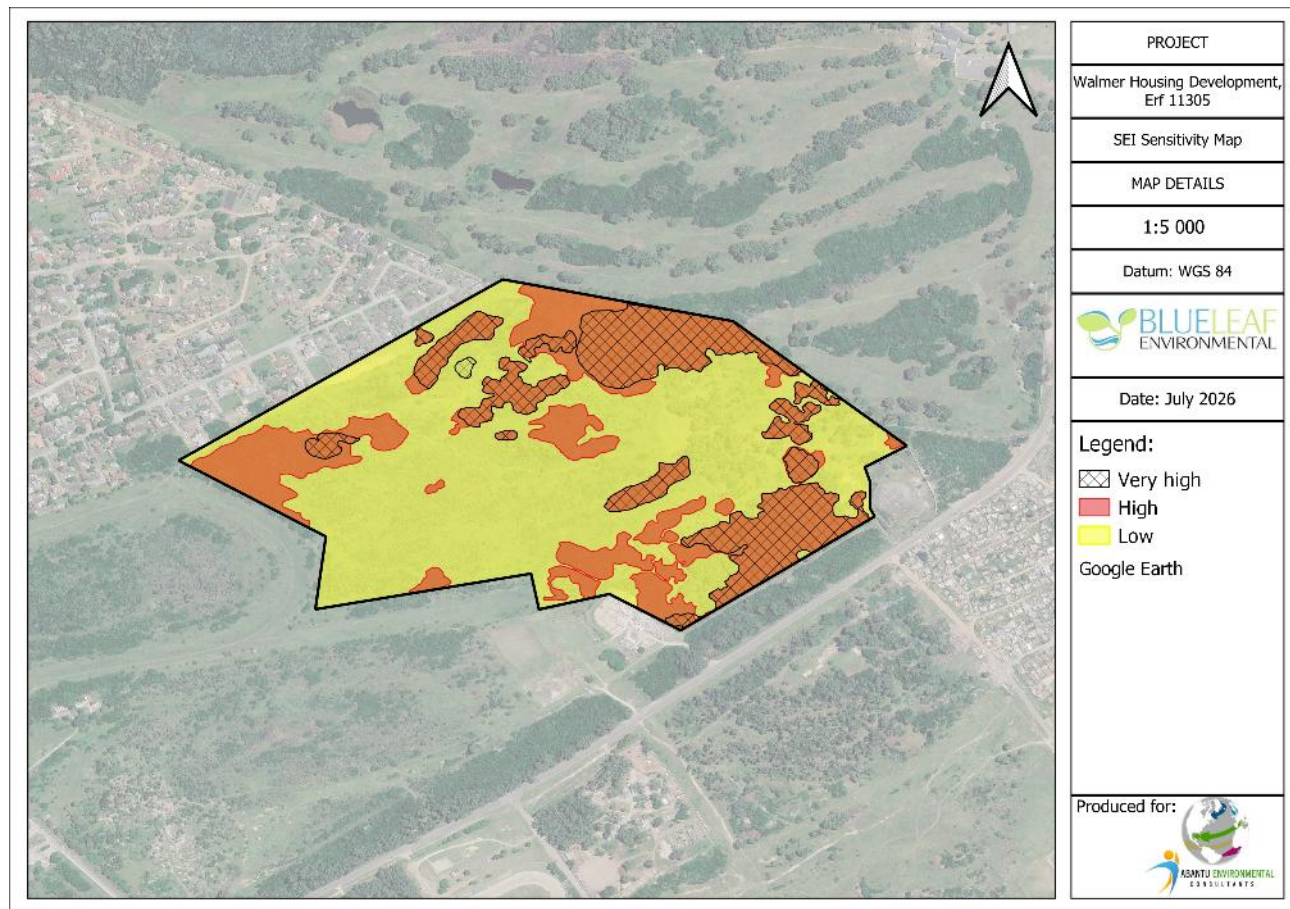


Figure 4.1: Site Ecological Importance sensitivity map for the terrestrial biodiversity theme for the proposed project site.

5. Impact assessment

The following issues were identified during the assessment of the proposed Walmer Housing Development (Erf 11305) in the Nelson Mandela Bay Municipality in the Eastern Cape Province.

5.1. Identified impacts

The following ecological issues were identified during the assessment of the proposed housing development:

Number	Activity causing impact (Issue)	Description of impact
1	Non-compliance with existing legislation	1.1. Legal compliance Non-compliance with ecological laws and policies of South Africa could lead to unnecessary delays in establishment activities, and potentially criminal cases, based on the severity of the non-compliance, being brought against the proponent and his/her contractors.
2	Vegetation clearing for development	2.1. Loss of natural vegetation Clearing will result in the loss of approx. 43.73 ha of vegetation during the construction phase of development.
		2.2. Loss of plant SCC Clearing may result in the loss of plant SCC (if present on site).
		2.3. Loss of animal SCC Clearing may result in the loss of animal SCC (if present on site).
		2.3. Spread of alien and invasive plant species Construction and the removal of natural vegetation will increase the risk of proliferation of alien and invasive plant species in open space areas on site, as well as into adjacent properties. A number of AIS are present on site.
3	Loss of biodiversity	3.1. Loss of biodiversity value Clearing will not result in the loss of CBA or ESA-valued areas (NMBM Bioregional Plan, 2015) during construction, as such areas are not present on site. However, the remnant, partially-intact patches of forest/thicket vegetation present on site are characteristic of Sardinia Forest Thicket (VegMap, 2024; NBA, 2025) – listed as a Critically Endangered ecosystem – and Sardinia Bay Forest Thicket (NMBM Bioregional Plan, 2015) – listed as a Vulnerable ecosystem. Approx. 7.6 ha (17%) of vegetation, including the greatest portion of forest/thicket patches (currently covering approx. 15.95 ha or 36% of the total project area), will remain as green open space (Open Space Zone 3: conservation usage) once construction is completed. The clearing of a large area of green open space, which is still used by birds, insects, small mammals etc., will reduce ecological connectivity and functioning in the landscape.
		3.2. Habitat fragmentation The housing development will create barriers within the landscape, fragmenting remaining stands of natural vegetation and hindering the movement of animals and dispersal of pollen/seeds. This can lead to isolated populations that are more vulnerable to genetic drift, inbreeding and extinction.

All impacts identified above were assessed as per the assessment methodology described in Section 2.7 of this report. Each impact was described based on how it would affect the project site during each specific phase of the project, namely Planning and Design, Construction and Operational Phases.

Issue 1:	Non-compliance with existing legislation
Consequence of Issue	Non-compliance with ecological laws and policies of South Africa could lead to unnecessary delays in establishment activities, and potentially criminal cases, based on the severity of the non-compliance, being brought against the proponent and his/her contractors. Permits will be required for the removal of any protected plant and animal species for some areas on the farm.
Number of impacts identified associated with this issue	Only 1 (Impact 1.1)

Impact 1.1: Legal compliance
Phase of expansion: Planning and Design Phase/Construction Phase

Nature of impact	Non-compliance with ecological laws and policies of South Africa could lead to unnecessary delays in establishment activities, and potentially criminal cases, based on the severity of the non-compliance, being brought against the proponent and his/her contractors.		
Cumulative impact	None		
Indirect impacts	None		
Residual impacts	None		
Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	4	2	Only during construction phase.
Extent of impact	4	1	Provincial approval will be required.
Intensity of impact	4	2	Legislated approval is required to impact on any protected plants, animals, or environments.
Severity	12	5	Duration + extent + intensity
Probability of impact occurring	5	2	Impact will occur on commencement of construction.
Frequency	5	1	Required prior to commencement of activities.
Incidence	10	3	Frequency + Probability
Degree of reversibility	High		Permits and authorisations may be required.
Irreplaceability	Low		No resource will be lost.
Mitigations	Mitigatory potential		Recommended mitigations
	High		- All relevant plant and animal removal/relocation permits must be obtained from the competent authorities to remove any protected plant and animal species. - Biodiversity offset agreements (if required) must be obtained before commencement of any clearing on site.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	High negative (120)		Very low negative (15)

Issue 2:	Vegetation clearing for development
Consequence of issue	Clearing of vegetation will result in a range of issues including increasing the risk of losing species of sensitive plants and animals (if present on site), reducing indigenous vegetation types, and increasing the risk of alien vegetation spreading.
Number of impacts	4 (Impacts 2.1 to 2.4)

Impact 2.1: Loss of natural vegetation	
Phase of expansion: Construction Phase	
Nature of impact	Clearing will result in the loss of approx. 43.73 ha of vegetation during construction of the developments.
Cumulative impact	Loss of biodiversity, habitats and ecological functioning in the general surrounding area.
Indirect impacts	Loss of biodiversity, habitats and ecological functioning in the general surrounding area.
Residual impacts	Loss of vegetation (green open space) within the general surrounding area.

Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	5	5	Clearing will result in the permanent loss of 43.73 ha of vegetation within the project site.
Extent of impact	2	1	Vegetation within the construction footprint only will be removed. The greatest portion of forest/thornveld patches (7.6 ha or 17% of the total project area), classed as a Critically Endangered/Vulnerable ecosystem, will remain within the project site. Another 12.9 ha (29%) of the site will remain as green open space (grassed or maintained areas).
Intensity of impact	4	2	Ecological processes may be impacted.
Severity	11	8	Duration + extent + intensity
Probability of impact occurring	5	4	Impact will occur on commencement of clearing.
Frequency	5	4	Once-off occurrence but will be permanent.
Incidence	10	8	Frequency + Probability
Degree of reversibility	High		Impact can be reversed.
Irreplaceability	Low		Vegetation will be removed in the footprint. Natural vegetation remains in the surrounding landscapes.
Mitigations	Mitigatory potential		Recommended mitigations
	High		- The construction footprint must be surveyed and demarcated prior to commencement of clearing. - No activities (temporary or permanent) will be allowed outside the approved footprint. - Permits must be obtained from Forestry to impact in any of the delineated forest patches.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	High negative (110)		Medium negative (64)

Impact 2.2: Loss of plant SCC	
Phase of expansion: Construction Phase	
Nature of impact	Clearing may result in the loss of non-identified plant SCC and other protected plant species. No plant SCC were identified on site.
Cumulative impact	Reduction in individual plant SCC and protected plant species numbers.
Indirect impacts	Loss in genetic variability within a specific plant SCC and protected species.
Residual impacts	Localized extinction of plant SCC and protected species.

Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	5	1	Removal of SCC will only occur during the initial stages of clearing.
Extent of impact	2	1	Only SCC within the footprint will be removed.
Intensity of impact	3	1	Genetic viability will be permanently lost.
Severity	10	3	Duration + extent + intensity
Probability of impact occurring	3	1	Impact will occur on commencement of construction.
Frequency	3	1	Throughout construction phase.
Incidence	6	2	Frequency + Probability
Degree of reversibility	Low negative		Impact can be reversed through mitigation.
Irreplaceability	Moderate		Plant species (and genetics) will be permanently lost.
Mitigations	Mitigatory potential		Recommended mitigations
	Medium		- No plant SCC were observed. Other plants protected by provincial legislation do occur. - All relevant plant permits must be obtained prior to commencement of clearing. - A qualified botanist must be appointed to oversee all protected plant removal. - Removed plants must be planted in a similar environment nearby. Alternatively, plants can be kept in an onsite nursery or donated to a local rehabilitation nursery. - Some of the plants may be used during rehabilitation of the site after development. - No plant harvesting by construction staff will be allowed.
Significance of impact (Severity x Incidence)	Pre-mitigation significance Medium negative (60)		Post-mitigation significance Very low negative (6)

Impact 2.3: Loss of animal SCC	
Phase of expansion: Construction Phase	
Nature of impact	Clearing may result in the loss of non-identified animal SCC and other protected animal species. No animal SCC were identified on site.
Cumulative impact	Reduction in individual protected animal species numbers.
Indirect impacts	Loss in genetic variability within a specific protected species.
Residual impacts	Reduction in individual protected species numbers.

Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	5	1	Removal of SCC will only occur during the initial stages of clearing. Some animals may be found during construction and operational phases e.g. small mammals (moles, rats) and reptiles.
Extent of impact	2	1	Animal habitats will be impacted.
Intensity of impact	2	1	Habitats will be lost.
Severity	9	3	Duration + extent + intensity
Probability of impact occurring	2	1	Impact will occur on commencement of construction.
Frequency	3	1	Throughout construction and operational phases.
Incidence	5	2	Frequency + Probability
Degree of reversibility	Low negative		Impact can be reduced through mitigation.
Irreplaceability	Moderate		Animal species (and genetics) will be permanently lost.
Mitigations	Mitigatory potential		Recommended mitigations
	Medium		- No animal SCC were identified. - Animals protected by provincial legislation may occur. These include local reptiles (snakes, lizards, geckos, tortoises and chameleons). All reptiles in the Eastern Cape are protected. - Permits (if required) must be obtained to relocate any protected animals identified. - Animals must only be relocated into the surrounding habitats. - Animals must be relocated by a trained and qualified handler as some of them may be venomous. - No poaching by construction staff will be allowed. - No killing of wild animals will be allowed.
Significance of impact (Severity x Incidence)	Pre-mitigation significance Low negative (45)		Post-mitigation significance Very low negative (6)

Impact 2.4: Spread of alien and invasive plant species	
Phase of expansion: Construction and Operational Phases	
Nature of impact	Construction and vegetation clearing will increase the risk of alien plant species invasion.
Cumulative impact	Increase in regional spread of alien plants.
Indirect impacts	Spread of alien vegetation into surrounding areas.
Residual impacts	Decreased risk of alien vegetation occurrence.

Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	5	2	Clearing will occur in the initial stages of construction.
Extent of impact	2	1	Only the construction footprint will be impacted.
Intensity of impact	3	1	All cleared areas have a risk of alien vegetation growth.
Severity	10	4	Duration + extent + intensity
Probability of impact occurring	4	1	Alien vegetation will grow on cleared areas and soil stockpiles.
Frequency	5	3	Impact will occur throughout construction and operational phases.
Incidence	9	4	Frequency + Probability
Degree of reversibility	High		Impact can be managed throughout mitigation.
Irreplaceability	Low		All alien vegetation must be repeatedly removed within footprint during construction and operational phases.
Mitigations	Mitigatory potential		Recommended mitigations
	Medium		- Develop and implement an Alien Vegetation Management Plan to mitigate the establishment and spread of undesirable alien plant species during vegetation clearing and construction and operational phases. - Site must be continuously monitored and managed for alien vegetation growth. - Removal must occur through appropriate methods such as hand pulling, application of chemicals, cutting, etc. in accordance with the NEMBA: Alien and Invasive Species Regulations.
Significance of impact (Severity x Incidence)	Pre-mitigation significance Medium negative (90)		Post-mitigation significance Very low negative (16)

Issue 3:	Loss of biodiversity
Consequence of issue	Clearing of vegetation and functioning of the site as a residential, business and recreational area, may result in a further loss in biodiversity value and habitat fragmentation, resulting in the loss of surrounding sensitive vegetation units or habitats (forest/thicket and grassy thornveld).
Number of impacts	2 (Impacts 3.1 and 3.2)

Impact 3.1: Loss of biodiversity value	
Phase of expansion: Construction and Operational Phases	
Nature of impact	The development will create barriers within the landscape, fragmenting remaining natural habitats and hindering the movement of animals and dispersal of seeds. This can lead to isolated populations that are more vulnerable to genetic drift, inbreeding, and extinction.
Cumulative impact	Contributing to the localized loss of sensitive habitats.
Indirect impacts	Loss of local sensitive plant and animal species.
Residual impacts	Loss of sensitive habitats.

Classification of impact	Before mitigating	After mitigating	Consequence of Impact (pre-mitigation only)
Duration of impact	5	4	Impact will be long term.
Extent of impact	2	1	Site and surrounding areas will be impacted.
Intensity of impact	5	3	Impact on ecological functions.
Severity	12	8	Duration + extent + intensity
Probability of impact occurring	5	3	Impact will occur.
Frequency	5	3	Throughout construction phase (continuously).
Incidence	10	6	Frequency + Probability
Degree of reversibility	High		Impact can be avoided by implementing the proposed mitigation measures.
Irreplaceability	High		Functionality changes will be temporary.
Mitigations	Mitigatory potential		Recommended mitigations
	High		– All areas outside the demarcated construction footprint will be considered as No-Go areas. – Forest patches must be avoided. Permits must be obtained from Forestry to impact in any of the delineated forest patches.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	High negative (120)		Medium negative (48)

Impact 3.2.: Habitat fragmentation	
Phase of expansion: Construction and Operational Phases	
Nature of impact	Negative. The development may create barriers within the landscape, fragmenting remaining stands of forest/thicket and thornveld vegetation and hindering the movement of animals and dispersal of pollen/seeds. This can lead to isolated populations of plant and/or animal SCC that are more vulnerable to genetic drift, inbreeding and extinction.
Cumulative impact	Increase in fragmentation of stands of natural vegetation in the general area.
Indirect impacts	Increased edge effects due to changes in microclimate, increased susceptibility to AIS, and limitation of gene flow between populations.
Residual impacts	Loss of plant and animal diversity, including SCC, and increased threat of plant and animal extinctions.

Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	5	4	Fragmentation of habitat will be long term.
Extent of impact	2	1	Will impact the development footprint and surrounding areas.
Intensity of impact	3	2	Impact will occur on commencement of construction.
Severity	10	7	Duration + extent + intensity
Probability of impact occurring	4	3	Impact will occur on commencement of construction.
Frequency	5	3	Impact will occur on commencement of construction and will be long term.
Incidence	10	6	Frequency + Probability
Degree of reversibility	Low		Impact can be partially reversed through mitigation measures.
Irreplaceability	High		Vegetation will be permanently lost.
Mitigations	Mitigatory potential		Recommended mitigations
	Low		No areas of forest/thicket vegetation occurring outside the areas designated 'Open Space Zone 3 (conservation usage)' in the site layout plan/rezoning and subdivision plan should be lost because of the clearing.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	High negative (100)		Low negative (42)

6. Specialist opinion

BlueLeaf has been appointed by Abantu Environmental Services (Pty) Ltd to undertake a Terrestrial Biodiversity, Plant and Animal Species Specialist Assessment as part of the EIA process being conducted by Abantu Environmental Services for the proposed establishment of the Walmer Housing Development on Erf 11305, west of Walmer Township, in Port Elizabeth/Gqeberha in the NMBM in the Eastern Cape Province. The contents of this specialist report comply with the legislated requirements as described in the following environmental theme and associated specialist assessment protocols as listed in the project Screening Tool Report:

1. Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Terrestrial Biodiversity.
2. Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Terrestrial Plant Species.
3. Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Terrestrial Animal Species.

Based on the outcome of the Site Sensitivity Verification Process, the three themes were as follows:

1. **Plant Species theme** – Medium sensitivity
2. **Animal Species theme** – Low sensitivity
3. **Terrestrial Biodiversity theme** – Very high sensitivity.

Based on the site sensitivity verification outcome for these themes, the plant and animal species themes were therefore assessed as Compliance Statements. The terrestrial biodiversity theme was assessed as a full Terrestrial Biodiversity Impact Assessment. The protocols for each theme (in GN.R. 1150 of 2020) were followed.

Three fine-scale vegetation units were identified on site: invaded grassland, forest-thicket patches and built-up and excavated areas.

6.1. Alternatives

No site alternatives or layout alternatives were proposed. The proposed development is NOT considered as fatally flawed provided that all mitigation measures provided in this report are implemented.

6.2. Cumulative impacts

In terms of Environmental Impact Assessment, Cumulative Impact is defined as:

“Means the past, current and reasonably foreseeable future impact of an activity, considered together with the impact of activities associated with that activity, that in itself may not be significant, but may become significant when added to the existing and reasonably foreseeable impacts eventuating from similar or diverse activities”.

The following cumulative impacts were identified:

1. **Loss of natural vegetation and plant and animal species.** Vegetation within the proposed direct development footprint (total 43.73 ha) is mostly secondary and degraded/transformed in nature i.e. invaded grassland and remnant, partially intact forest/thicket patches with the potential to host plant and animal SCC (though not observed at the time of the site survey). Though no longer representative of the original Critically Endangered or Vulnerable vegetation/ecosystem types mapped for the site i.e. Sardinia Forest Thicket (SANBI VegMap, 2018/2024) or Sardinia Bay Forest Thicket (NMBM Bioregional Plan, 2015), respectively – these areas of ‘green open space’ do serve as corridors for the movement/browsing of animals and the transfer of the genetic material of plants. All forest patches must be avoided.
2. **Increased risk of alien invasive plants spreading to surrounding areas** because of vegetation clearing. This impact can be easily managed through the development and implementation of an Alien Vegetation Management Plan. It is important to note that this plan must be implemented during both the construction and operational phases of the proposed development.

6.3. Levels of acceptable change

The proposed development will not result in altered ecological services and functionality, provided that all mitigation measures proposed in this report are implemented.

6.4. Levels to be avoided

Due to the conversion of a vegetated area to hard and impenetrable surfaces, on steep or sandy slopes in places, the proposed development may result in the presence of a larger volume of runoff, with an increased risk of poor stormwater management and soil erosion on site, as well in surrounding areas. Provided that all mitigation measures proposed in this report are implemented, these risks (as identified in this report) are considered as acceptable changes to the local environment. All forest patches must be avoided.

6.5. Current impacts

The following impacts are currently occurring on site and will be increased/exacerbated by the proposed development:

- Historical activities like vegetation clearing and habitat fragmentation through the construction of roads, residential and commercial/business areas.
- The site is currently experiencing the impacts of habitat fragmentation, with a number of alien invasive plants identified on site. Left unmitigated, alien invasive species may spread and displace indigenous plant species. Plant and animal diversity may be further reduced.

6.6. General rehabilitation measures

The following general rehabilitation measures are recommended for inclusion into the EMPr:

- Implement an eradication/control programme for early intervention if invasive species are detected, so that their spread to surrounding natural ecosystems can be prevented.

- Rehabilitate/stabilise disturbed areas as quickly as possible to reduce the area where invasive species would be at a strong advantage and most easily able to establish.
- Implement a monitoring programme to detect alien invasive species early, before they become established and, in the case of weeds, before the release of seeds.

6.7. Biodiversity offsetting

As the areas to be impacted by the proposed housing development have largely been cleared of the original indigenous vegetation most likely present on site before human settlement i.e. AT48 Sardinia Forest Thicket (SANBI VegMap, 2018/2024) or Sardinia Bay Forest Thicket (NMBM Final Bioregional Plan (2015)), vegetation, as mapped for the site and surrounding areas – vegetation present within the proposed development footprint is largely degraded, transformed and secondary in nature. The areas to be impacted consist of largely of secondary grassland (previously cleared areas) and alien and invasive Australian Acacias (Rooikrans and Port Jackson Willow). Vegetation is, therefore, no longer representative of the original and Critically Endangered/Vulnerable vegetation/ecosystem type(s) present before development of the project site.

The only remnant, continuous and partially intact patches of indigenous vegetation present are the forest/thornveld vegetation patches present in the northern, northeastern and southeastern portions of the site, which will largely not be developed. 17% (7.6 ha) out of the total estimated 36% (15.95 ha) remnant forest/thicket vegetation will remain as areas zoned Open Space Zone 3 (conservation usage) once construction is completed (in the total 43.73 ha development footprint). All forest patches as shown in the Sensitivity Map must be avoided.

No biodiversity offset is, therefore, required.

6.8. Specialist opinion

The proposed Walmer Housing Development is NOT considered to be Fatally Flawed, and no components of the proposed development have been identified as flawed. No site or layout alternatives are proposed. Biodiversity offsetting is not required.

The ecological impacts of all aspects for the proposed development were assessed and considered to be acceptable, provided that all mitigation measures provided in this report are implemented.

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8. Appendix A – Plant Species List

Family	Scientific name	Red List (2024)	PNCO (1974)	Exotic/AIS	Observed (Y/N)	Potentially present
Acanthaceae	Hypoestes forskoolii (Vahl) R.Br.	LC			Y	
Aizoaceae	Carpobrotus deliciosus (L.Bolus) L.Bolus	LC	X Sched 4		Y	
Aizoaceae	Mesembryanthemum aitonis Jacq.	LC	X Sched 4		Y	
Amaranthaceae	Chenopodium murale L. var. murale	NE		X	Y	
Amaranthaceae	Chenopodium album L.	NE		X	Y	
Amoryllidaceae	Brunsvigia gregaria R.A.Dyer	LC	X Sched 4		Y	
Anacardiaceae	Schinus terebinthifolius Raddi	NE		X Cat 1b	Y	
Anacardiaceae	Harpephyllum afrum Bernh. ex Krauss	LC			Y	
Anacardiaceae	Searsia crenata (Thunb.) Moffett	LC			Y	
Anacardiaceae	Searsia glauca (Thunb.) Moffett	LC			Y	
Anacardiaceae	Searsia incisa (L.f.) F.A.Barkley var. effusa (C.Presl) Moffett	LC			N	X
Anacardiaceae	Searsia pterota (C.Presl) Moffett	LC			N	X
Apocynaceae	Carissa bispinosa (L.) Desf. ex Brenan	LC			Y	
Apocynaceae	Cynanchum obtusifolium L.f.	LC			Y	
Apocynaceae	Cynanchum natalitium Schltr.	LC			N	X
Apocynaceae	Cynanchum viminalis (L.) Bassi	LC			N	X
Araliaceae	Cussonia thyrsiflora Thunb.	LC			N	X
Asparagaceae	Agave sisalana Perrine	NE		X Cat 2	Y	
Asparagaceae	Asparagus asparagoides (L.) Druce	LC			Y	
Asparagaceae	Asparagus aethiopicus L.	LC			Y	
Asphodelaceae	Trachyandra ciliata (L.f.) Kunth	LC			Y	
Asphodelaceae	Aloe africana Mill.	LC	X Sched 4		N	X
Asteraceae	Arctotheca prostrata (Salisb.) Britten	LC			Y	
Asteraceae	Cotula discolor (DC.) J.C.Manning & Mucina	LC			Y	
Asteraceae	Erigeron sp.	NE		X	Y	
Asteraceae	Helichrysum aureum (Houtt.) Merr.	LC			Y	

Family	Scientific name	Red List (2024)	PNCO (1974)	Exotic/AIS	Observed (Y/N)	Potentially present
Asteraceae	Helichrysum cymosum (L.) D.Don	LC			Y	
Asteraceae	Hypochaeris radicata L.	NE		X	Y	
Asteraceae	Senecio glutinosus Thunb.	LC			Y	
Asteraceae	Senecio pterophorus DC.	LC			Y	
Asteraceae	Senecio inaequidens DC.	LC			Y	
Asteraceae	Felicia erigeroides DC.	LC			Y	
Asteraceae	Senecio ilicifolius L.	LC			Y	
Asteraceae	Lactuca inermis Forssk.	LC			Y	
Asteraceae	Tarchonanthus littoralis P.P.J.Herman	LC			N	X
Bignoniaceae	Tecomaria capensis (Thunb.) Spach	LC			Y	
Brassicaceae	Lepidium africanum (Burm.f.) DC. subsp. africanum	LC			Y	
Brassicaceae	Rapistrum rugosum (L.) All.	NE		X	Y	
Casuarinaceae	Casuarina sp.	NE		X Cat 2	Y	
Celastraceae	Mystroxydon aethiopicum (Thunb.) Loes. subsp. aethiopicum	LC			Y	
Celastraceae	Putterlickia pyracantha (L.) Szyszyl.	LC			Y	
Celastraceae	Pterocelastrus tricuspidatus (Lam.) Walp.	LC			N	X
Celastraceae	Cassine peragua L. subsp. peragua	LC			N	X
Celastraceae	Gymnosporia buxifolia (L.) Szyszyl.	LC			N	X
Celastraceae	Gymnosporia capitata (E.Mey. ex Sond.) Loes.	LC			N	X
Celastraceae	Maytenus procumbens (L.f.) Loes.	LC			N	X
Celastraceae	Robsonodendron maritimum (Bolus) R.H.Archer	LC			N	X
Colchicaceae	Colchicum eucomoides (Jacq.) J.C.Manning & Vinn.	LC			Y	
Crassulaceae	Cotyledon velutina Hook.f.	LC			N	X
Cucurbitaceae	Zehneria scabra (L.f.) Sond.	LC			Y	
Ebenaceae	Euclea racemosa Murray	LC			N	X
Euphorbiaceae	Euphorbia helioscopia L.	NE		X	Y	
Euphorbiaceae	Ricinus communis L. var. communis	NE		X Cat 2	Y	
Fabaceae	Acacia cyclops A.Cunn. ex G.Don	NE		X Cat 1b	Y	
Fabaceae	Acacia saligna (Labill.) H.L.Wendl.	NE		X Cat 1b	Y	
Fabaceae	Vachellia karroo (Hayne) Banfi & Gallaso	LC			Y	

Family	Scientific name	Red List (2024)	PNCO (1974)	Exotic/AIS	Observed (Y/N)	Potentially present
Fabaceae	Lessertia stenoloba E.Mey.	LC			Y	
Fabaceae	Melilotus indicus (L.) All.	NE		X	Y	
Fabaceae	Schotia afra (L.) Thunb.	LC			Y	
Fabaceae	Lessertia frutescens (L.) Goldblatt & J.C.Manning	LC			Y	
Fabaceae	Vicia sp.	NE			Y	
Fabaceae	Erythrina afra Thunb.	LC			N	X
Fumariaceae	Fumaria muralis Sond. ex W.D.J.Koch subsp. muralis	NE		X	Y	
Hyacinthaceae	Veltheimia bracteata Harv. ex Baker	LC	X Sched 4		N	X
Iridaceae	Moraea britteniae (L.Bolus) Goldblatt	LC	X Sched 4		Y	
Lamiaceae	Leonotis ocymifolia (Burm.f.) Iwarsson	LC			Y	
Malvaceae	Malva parviflora L. var. parviflora	NE		X	Y	
Malvaceae	Abutilon sonneratianum (Cav.) Sweet	LC			N	X
Malvaceae	Grewia occidentalis L. var. occidentalis	LC			N	X
Meliaceae	Melia azedarach L.	NE		X Cat 1b	Y	
Meliaceae	Ekebergia capensis Sparrm.	LC			Y	
Menispermaceae	Cissampelos capensis L.f.	LC			Y	
Myrtaceae	Eucalyptus sp.	NE		X Cat 2	Y	
Oleaceae	Olea capensis L.	LC			N	X
Oleaceae	Olea exasperata Jacq.	LC			N	X
Oxalidaceae	Oxalis sp.	LC			Y	
Pittosporaceae	Pittosporum viridiflorum Sims	LC	NFA Protected Tree		N	X
Plantaginaceae	Plantago lanceolata L.	LC			Y	
Plumbaginaceae	Plumbago auriculata Lam.	LC			N	X
Poaceae	Cynodon dactylon (L.) Pers.	LC			Y	
Poaceae	Pennisetum clandestinum Hochst. ex Chiov.	NE		X	Y	
Poaceae	Sporobolus africanus (Poir.) Robyns & Tournay	LC			Y	
Poaceae	Stenotaphrum secundatum (Walter) Kuntze	LC			Y	
Poaceae	Panicum deustum Thunb.	LC			Y	
Poaceae	Stenotaphrum secundatum (Walter) Kuntze	LC			Y	

Family	Scientific name	Red List (2024)	PNCO (1974)	Exotic/AIS	Observed (Y/N)	Potentially present
Poaceae	Eragrostis sp.	LC			Y	
Polygonaceae	Rumex sagittatus Thunb.	LC			Y	
Polygonaceae	Emex australis Steinh.	LC			Y	
Ranunculaceae	Clematis brachiata Thunb.	LC			Y	
Rhamnaceae	Scutia myrtina (Burm.f.) Kurz	LC			Y	
Rosaceae	Rubus pinnatus Willd.	LC			Y	
Rubiaceae	Galium capense Thunb. subsp. capense	LC			Y	
Rubiaceae	Anthospermum aethiopicum L.	LC			Y	
Rutaceae	Clausena anisata (Willd.) Hook.f. ex Benth. var. anisata	LC	X Sched 4		Y	
Rutaceae	Vepris lanceolata (Lam.) G.Don	LC	X Sched 4		N	X
Rutaceae	Zanthoxylum capense (Thunb.) Harv.	LC	X Sched 4		N	X
Salvadoraceae	Azima tetracantha Lam.	LC			N	X
Sapotaceae	Sideroxylon inerme L. subsp. inerme	LC	NFA Protected Tree		Y	
Scrophulariaceae	Chaenostoma campanulatum Benth.	LC			Y	
Scrophulariaceae	Hebenstretia integrifolia L.	LC			Y	
Solanaceae	Cestrum laevigatum Schltld.	NE		X Cat 1b	Y	
Solanaceae	Datura stramonium L.	NE		X Cat 1b	Y	
Solanaceae	Solanum africanum Mill.	LC			Y	
Solanaceae	Solanum sisymbriifolium Lam.	NE		X Cat 1b	Y	
Solanaceae	Physalis peruviana L.	NE		X	Y	
Thymelaceae	Passerina corymbosa Eckl. ex C.H.Wright	LC			Y	
Verbenaceae	Lantana camara L.	NE		X Cat 1b	Y	
Vitaceae	Cyphostemma cirrhosum (Thunb.) Desc. ex Wild & R.B.Drumm. subsp. cirrhosum	LC			Y	
Vitaceae	Rhoicissus tridentata (L.f.) Wild & R.B.Drumm.	LC			Y	
Vitaceae	Rhoicissus digitata (L.f.) Gilg & M.Brandt	LC			N	X
Zygophyllaceae	Roepera maritima (Eckl. & Zeyh.) Beier & Thulin	LC			Y	
Zygophyllaceae	Roepera morgsana (L.) Beier & Thulin	LC			N	X